



STARBOOKS

Whiz Challenge

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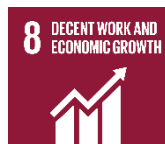
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Software Requirements Specification

for

STARBOOKS Whiz Challenge

Version 1.0

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Contents

CONTENTS	II
REVISIONS	III
1 INTRODUCTION	1
1.1 DOCUMENT PURPOSE.....	1
1.2 PRODUCT SCOPE.....	1
1.3 INTENDED AUDIENCE AND DOCUMENT OVERVIEW.....	1
1.4 DEFINITIONS, ACRONYMS AND ABBREVIATIONS.....	2
1.5 DOCUMENT CONVENTIONS	3
2 OVERALL DESCRIPTION	4
2.1 PRODUCT OVERVIEW.....	4
2.2 PRODUCT FUNCTIONALITY	4
2.3 DESIGN AND IMPLEMENTATION CONSTRAINTS.....	6
2.4 ASSUMPTIONS AND DEPENDENCIES	7
3 SPECIFIC REQUIREMENTS	9
3.1 EXTERNAL INTERFACE REQUIREMENTS	9
3.2 FUNCTIONAL REQUIREMENTS	13
3.3 USE CASE MODEL	21
4 OTHER NON-FUNCTIONAL REQUIREMENTS	22
4.1 PERFORMANCE REQUIREMENTS.....	22
4.2 SAFETY AND SECURITY REQUIREMENTS	24
4.3 SOFTWARE QUALITY ATTRIBUTES	26
5 OTHER REQUIREMENTS	30
5.1 DATAFLOW DIAGRAMS.....	30
5.2 ACTIVITY DIAGRAMS WITH SWIM-LANES	39
5.3 SEQUENCE DIAGRAMS.....	46
5.4 DEPLOYMENT DIAGRAM.....	54
APPENDIX A – ENTITY RELATIONSHIP DIAGRAM	56
APPENDIX B - GROUP LOG	57
APPENDIX C – TEST CASES	68

Revisions

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1 Introduction

1.1 Document Purpose

This Software Requirements Specification (SRS) document identifies and describes the software requirements for the STARBOOKS Whiz Challenge Version 1.0, an interactive, offline educational application developed for the Department of Science and Technology – Science and Technology Information Institute (DOST-STII). This document serves as a comprehensive guide for all stakeholders, including developers, project managers, testers, DOST-STII administrators, and other parties involved in the development, deployment, and maintenance of the system.

The scope of this SRS encompasses the complete STARBOOKS Whiz Challenge system, covering both the client-side gamified learning platform and the web-based administrative dashboard. This includes all educational mini-games (Whiz Challenge, Whiz Memory, Whiz Puzzle, and Whiz Battle), the reward and leaderboard systems, and the administrative tools for user and content management and game configurations. This document details the functional and non-functional requirements, system constraints, interface specifications, and UML diagrams necessary to guide the successful implementation of the STARBOOKS Whiz Challenge platform.

1.2 Product Scope

The STARBOOKS Whiz Challenge is a gamified educational platform designed to complement the existing DOST-STII STARBOOKS project by making Science, Technology, Engineering, and Mathematics (STEM) learning more interactive and enjoyable. The system is primarily targeted at students—ranging from elementary to senior high school—who visit DOST events, science fairs, and public exhibits, though it remains accessible to all event visitors. The administrative dashboard is exclusively designed for DOST-STII staff to manage content, monitor player engagement, and analyze educational impact through analytics.

This enhanced version addresses critical limitations observed in the current Whiz Challenge application, including limited replayability, lack of diverse learning strategies, insufficient performance tracking, and the need for a more modern user experience. By introducing multiple game modes (Whiz Challenge, Whiz Memory, Whiz Puzzle, and Whiz Battle), the system accommodates different learning styles and promotes sustained engagement through varied gameplay. The integration of a rewards system with stars, badges, and leaderboards encourages continuous participation and healthy competition among learners. Key benefits include improved student retention and engagement in STEM topics, offline accessibility for areas with limited internet connectivity, data-driven insights for DOST-STII to measure educational impact and refine outreach strategies, and a visually appealing, modern interface that captures the attention of today's digital learners. Ultimately, the STARBOOKS Whiz Challenge aims to strengthen scientific literacy nationwide while supporting DOST-STII's mission to make quality STEM education accessible, engaging, and measurable.

1.3 Intended Audience and Document Overview

This Software Requirements Specification is intended for multiple audiences, each with specific interests in different aspects of the STARBOOKS Whiz Challenge project:

- **DOST-STII:** The primary stakeholder who will use this document to verify that the proposed system meets their educational objectives, operational needs, and strategic goals for promoting STEM literacy.
- **SoCIT Faculty:** Faculty members assessing the completeness, technical rigor, and feasibility of the requirements specification.
- **Future Maintainers:** Developers or technical staff who may be responsible for system maintenance, updates, and enhancements after initial deployment.

This document is organized into five main sections and three appendices. Section 1 (Introduction) provides an overview of the document's purpose, product scope, and organizational structure. Section 2 (Overall Description) presents a high-level view of the product, including its functionality, user characteristics, constraints, and dependencies. Section 3 (Specific Requirements) details the external interfaces, functional capabilities, and use case models that define system behavior. Section 4 (Other Non-Functional Requirements) addresses performance, safety, security, and quality attributes. Section 5 (Other Requirements) captures all other UML diagrams. Appendix A (ERD/Data Dictionary) defines all data elements and structures used throughout the system. Appendix B (Group Logs) documents the development team's progress and activities. Appendix C (Test Cases) provides detailed test scenarios for system validation.

1.4 Definitions, Acronyms and Abbreviations

- **AJAX Communication:** Asynchronous JavaScript and XML – A web development technique that allows data to be exchanged with a server and parts of a webpage to update without reloading the whole page.
- **API:** Application Programming Interface – A set of rules and tools that allow software applications to communicate with each other.
- **DFD:** Data Flow Diagram – A graphical representation that shows how data moves through a system, including inputs, processes, and outputs.
- **DOM Manipulation:** Document Object Model Manipulation – The process of dynamically changing the structure, style, or content of a webpage using programming (commonly JavaScript).
- **DOST-STII:** Department of Science and Technology – Science and Technology Information Institute – The information arm of the Philippines' Department of Science and Technology, responsible for science and tech communication.
- **GB:** Gigabyte – A unit of digital storage equal to about one billion bytes.
- **LAN:** Local Area Network – A computer network that connects devices within a limited area (like an office or school)
- **MVP:** Minimum Viable Product – The simplest version of a product that can be released to test core functionality and gather user feedback before further development.
- **NIC:** Network Interface Card – Hardware that connects a computer to a network, enabling communication over LAN or internet.
- **PC:** Personal Computer – A general-purpose computer designed for individual use.
- **QA:** Quality Assurance – A process that ensures products or services meet defined standards and requirements.
- **RAM:** Random Access Memory – Temporary computer memory used to store data and instructions while a system is running.
- **SDLC:** Software Development Life Cycle – A structured process for planning, creating, testing, and deploying software.
- **SRS:** Software Requirements Specification – A document that describes what a software system should do and how it should perform.
- **STEM:** Science, Technology, Engineering, and Mathematics – An academic and professional field grouping these disciplines, often emphasized in education and innovation.

- **TCP/IP:** Transmission Control Protocol/Internet Protocol – The fundamental communication protocols that define how data is transmitted over the internet
- **UML:** Unified Modeling Language – A standardized visual language used to design, specify, and document software systems, much like a blueprint for system architecture.

1.5 Document Conventions

Formatting Conventions

- Font: Arial, size 11 or 12, for all text to ensure readability.
- Spacing: Single-spaced text with 1-inch margins on all sides to maintain a clean layout.
- Section Titles: Bold and in a larger font size to clearly delineate different sections.

Naming Conventions

- Sections and Subsections: Numbered according to the IEEE style guide for consistency

2 Overall Description

2.1 Product Overview

The STARBOOKS Whiz Challenge Version 1.0 is a complete ground-up redevelopment of the existing DOST-STII STARBOOKS Whiz Challenge application. While the original application introduced gamified STEM learning through a solo quiz format, this new version addresses identified limitations by offering multiple game modes, enhanced engagement features, and comprehensive administrative tools. Developed as a standalone system, it operates independently of the current STARBOOKS infrastructure and is designed to demonstrate improved educational outcomes and user engagement.

The system architecture is designed for offline, LAN-based deployment to ensure accessibility in venues with limited or no internet connectivity. The solution consists of three primary components:

1. a web-based **game client** accessed through browsers on player PCs
2. a web-based **administrative dashboard** for DOST staff to manage content and users, monitor real-time player activity, and analyze performance metrics; and
3. a **centralized backend server** running Laravel with MongoDB for data management and game logic processing.

During DOST events, a dedicated server PC hosts the backend application and connects to multiple client PCs via a local area network (LAN) router. Players access the game through their web browsers, with all data—including player profiles, game progress, scores, and analytics—stored locally on the server. The administrative dashboard, also accessed via browser, communicates with the same server to provide real-time insights and content management capabilities.

2.2 Product Functionality

The STARBOOKS Whiz Challenge system provides the following major functions:

Player Management and Engagement

- User registration and login system for tracking individual player progress and maintaining personalized game experiences
- Player profile management with demographic information for analytics purposes

Game Modes and Gameplay

- **Whiz Challenge:** Solo quiz game mode where players answer questions to earn badges and rewards and climb the leaderboard
- **Whiz Memory Game:** Memory-matching game that reinforces learning through visual recall and pattern recognition
- **Whiz Puzzle:** Jigsaw puzzle-solving challenges that promote critical thinking and problem-solving skills
- **Whiz Battle:** Competitive 1v1 quiz showdown where two players compete head-to-head in real-time

Content Categorization and Difficulty

- Science and Mathematics subject categories covering diverse STEM topics
- Three difficulty levels (Easy, Average, Difficult) to accommodate learners of varying skill levels
- Configurable question pools for each category and difficulty combination

Rewards and Motivation System

- Badge and reward system that recognizes player achievements and milestones in Whiz Challenge and Whiz Battle
- Star-based scoring system that reflects player performance in Whiz Memory Match and Whiz Puzzle
- Reward claiming mechanism for players to redeem earned badges in Whiz Challenge and Whiz Battle
- Leaderboard rankings for Whiz Challenge and Whiz Battle to encourage healthy competition

Administrative Content Management

- Manage player accounts and administrative user lists
- Create, edit, and delete quiz questions across categories and difficulty levels
- Configure game parameters for Whiz Challenge and Whiz Battle including number of questions, score points, and time limits for each difficulty level
- Manage badge definitions and monitor reward claiming activities
- View and export leaderboard rankings for reporting and recognition purposes

Analytics and Reporting

Comprehensive game analytics dashboard with visual insights including:

- Total registered players and growth trends
- Most played game mode identification
- Average session duration tracking
- Gender distribution analysis (Male vs Female registered players)
- Age distribution demographics
- Geographic distribution (Players by region)
- Game mode preferences by gender
- Rewards distribution analysis by gender and difficulty level
- Game mode popularity segmented by age groups
- Export functionality for all analytics data for reporting purposes

System Operations

- Offline, LAN-based operation ensuring accessibility without internet connectivity
- Real-time data synchronization between player clients and central server
- Secure local data storage and retrieval using MongoDB database
- Multi-user concurrent access support for simultaneous gameplay during events

2.3 Design and Implementation Constraints

The development and implementation of the STARBOOKS Whiz Challenge system are subject to the following constraints and limitations:

Hardware and Performance Constraints

- The system must operate on standard Windows-based personal computers commonly available in DOST event venues, with typical specifications including Intel Core i3 or equivalent processors, minimum 4GB RAM, and at least 80GB available storage space.
- The server PC must support concurrent connections from multiple client machines without significant performance degradation.
- Client PCs must run modern web browsers (Chrome or Edge) with HTML5 and JavaScript support.

Technology and Platform Requirements

- **Frontend Development:** The game client interface must be developed using Flutter/Dart framework, compiled for web deployment to ensure cross-browser compatibility and responsive design.
- **Backend Development:** The server-side application must be built using the Laravel PHP framework running on Laragon development stack for local server management.
- **Database Management:** MongoDB must be used for all data storage, providing flexible schema design for game data, player profiles, and analytics.
- **Development Methodology:** The project must follow the Software Development Life Cycle (SDLC) using Agile/Scrum methodology for iterative development, with regular sprints and stakeholder feedback integration.
- **Design and Modeling:** System design must utilize the Unified Modeling Language (UML) for creating system models including use case diagrams, sequence diagrams, deployment diagram, etc.

Network and Connectivity Constraints

- The system must operate exclusively in an offline, Local Area Network (LAN) environment without requiring internet connectivity.
- Network infrastructure must support standard Ethernet connections through a local router.
- All client-server communication must occur within the isolated LAN to ensure data security and system reliability during events.

Security and Privacy Requirements

- The system must comply with the Philippine Data Privacy Act of 2012 (Republic Act No. 10173) regarding the collection, storage, and processing of personal information from players.
- Player data including names, ages, gender, and regional information must be stored securely with appropriate access controls.
- Administrative access must be protected through authentication mechanisms, with role-based access control separating player and admin functionalities.
- Sensitive data must not be transmitted outside the local network, maintaining data sovereignty within DOST-controlled infrastructure.

Integration and Interface Constraints

- The system operates as a standalone application with no integration requirements to existing DOST systems or external third-party services.
- All system components must communicate exclusively through RESTful API interfaces between the Laravel backend and Flutter frontend.
- The administrative dashboard and game client must be accessible through standard web browsers without requiring additional plugin installations.

Organizational and Maintenance Constraints

- Upon project completion, the complete system including source code, documentation (SRS and User Manual), and deployment instructions must be delivered to DOST-STII for their ownership and future maintenance.
- Code must be well-documented with inline comments and follow standard coding conventions to facilitate future maintenance by DOST-STII technical staff.
- The system must be designed for easy content updates, allowing non-technical DOST staff to manage questions, badges, and game configurations through the administrative interface.
- All documentation must be written in clear, professional English suitable for technical and non-technical stakeholders.

Regulatory and Standards Compliance

- The system must adhere to DOST-STII branding guidelines and educational content standards for science and mathematics materials.
- All educational content must be age-appropriate and aligned with the Department of Education's K-12 curriculum standards for STEM subjects.
- The system must support accessibility considerations, including readable fonts, appropriate color contrast, and intuitive navigation for diverse age groups.

2.4 Assumptions and Dependencies

The successful development, deployment, and operation of the STARBOOKS Whiz Challenge system are based on the following assumptions and dependencies:

Environmental and Infrastructure Assumptions

- Event venues will have reliable electrical power supply throughout the duration of DOST events, with sufficient power outlets to support multiple PC setups.
- Adequate physical space, tables, and seating arrangements will be available at event venues to accommodate the server PC, client PCs, and players.
- DOST-STII technical staff possess the necessary skills and knowledge to configure and maintain Local Area Network (LAN) infrastructure, including router setup, IP address management, and basic network troubleshooting.
- The venue environment will provide appropriate lighting and climate control to ensure comfortable gameplay experiences and proper equipment operation.

User Capability Assumptions

- Players (students and event visitors) possess basic computer literacy skills, including the ability to use a mouse, keyboard, and navigate web browsers.

- DOST-STII administrative staff have familiarity with web-based management interfaces and can perform routine content management tasks such as adding questions, updating configurations, and viewing analytics reports.
- All users (players and administrators) have sufficient English language proficiency to understand the game interface, instructions, and administrative controls.

Content and Educational Material Dependencies

- All educational content including quiz questions, correct answers, difficulty classifications, and subject categorizations will be provided by DOST-STII content specialists.
- The development team assumes that all questions provided by DOST-STII have been validated for scientific accuracy, age-appropriateness, and alignment with K-12 STEM curriculum standards.
- DOST-STII will be responsible for periodic content updates and quality assurance of educational materials post-deployment.
- Initial content database will be populated with a sufficient number of questions across all categories and difficulty levels before system launch.

Hardware and Equipment Dependencies

- DOST-STII will provide all necessary hardware including server PC, client PCs, LAN router, network cables, and peripherals (monitors, keyboards, mice) that meet the minimum specifications outlined in Section 2.3.
- All provided PCs will have compatible operating systems and up-to-date web browsers pre-installed.
- Hardware will be maintained in good working condition, with DOST-STII technical staff responsible for troubleshooting hardware failures, replacing defective components, and performing routine maintenance.

Technical Dependencies

- The system relies on the Flutter framework, Laravel framework, and MongoDB database remaining stable and supported throughout the development lifecycle and operational period.
- Standard web browser technologies will continue to be supported by major browsers without significant breaking changes.
- The Laragon development stack will provide a stable local server environment for Laravel and MongoDB deployment.

Organizational and Support Assumptions

- DOST-STII will assign dedicated technical staff with sufficient time and resources to learn system administration, perform routine maintenance, and provide on-site technical support during events.
- DOST-STII management will allocate appropriate budget for future hardware upgrades, replacements, or expansions as needed to support growing event attendance.
- DOST-STII will establish internal policies and procedures for data backup, system security, and incident response to ensure long-term system reliability.
- Post-deployment technical issues will be handled by DOST-STII staff, with the development team providing comprehensive documentation (SRS and User Manual) to facilitate knowledge transfer.

Deployment and Operational Assumptions

- The system will be deployed in controlled DOST event environments where network isolation can be maintained, minimizing security risks.
- Event schedules will allow adequate time for system setup, testing, and troubleshooting before player access begins.
- DOST-STII will implement appropriate physical security measures to prevent unauthorized access to server PCs and protect stored player data.
- Network bandwidth and server resources will be sufficient to handle peak loads during high-traffic periods at large events.

3 Specific Requirements

3.1 External Interface Requirements

3.1.1 User Interfaces

The STARBOOKS Whiz Challenge system provides two web-based interfaces accessed through standard browsers: a **player interface** for game participants and an **administrative dashboard** for DOST-STII staff.

Player Interface

The player interface features a colorful, child-friendly design with the STARBOOKS mascot and light blue backgrounds. Key screens include:

- a. Authentication Screens
 - Login screen with username/password fields and link to registration
 - Multi-step registration process collecting player information (username, password, school, age, avatar, category, sex, region, province, city)
 - Admin link in top-right corner for staff access
- b. Game Screens
 - Home navigation bar with STARBOOKS logo, Home button, Leaderboard button, Badge Collection, and player profile settings
 - Game mode configuration screens showing category selection (Science/Math) and three difficulty levels (Easy, Average, Difficult)
 - Whiz Challenge: Question display with timer, score counter, and four answer options in colored buttons
 - Whiz Memory: Card grid layout with timer, and move counter
 - Whiz Puzzle: Draggable jigsaw pieces with timer and move counter
 - Whiz Battle: Player matchup screen showing both avatars, usernames, and unique game code
- c. Leaderboard Screen
 - Table displaying rank, avatar, username, total rewards, rewards by difficulty level, and last claim timestamp

Administrative Dashboard

The admin interface uses a professional dark sidebar navigation with light blue content area for efficient management.

- a. Layout Components
 - Left sidebar with collapsible menu including: Analytics, Leaderboard, Users (Players/Admins), and Quiz Content (Questions/Difficulty)
 - Main content area displaying selected module
- b. Key Admin Screens
 - Analytics Dashboard: Displays summary metrics (total registered players, most played game mode, average session duration) alongside interactive visualization panels for demographic analysis, game mode preferences, and reward distribution statistics. All data can be filtered and exported for reporting purposes.
 - Leaderboard Management: Provides rankings with full player statistics, reward breakdowns by difficulty level, and export functionality for record-keeping.
 - Player Management: Complete player database with profile information, demographic categorization, and account management tools. Administrators can add new players, edit existing profiles, and manage the reward claiming system.
 - Admin Management: User administration panel for managing DOST-STII staff accounts, including the ability to add new administrators, and edit admin profiles.
 - Quiz Content Management: Comprehensive question bank organized by subject category (Science/Math) and difficulty level. Features include adding new questions and editing existing content.
 - Difficulty Management: Configuration interface displaying three difficulty tiers (Easy, Average, Difficult) with editable parameters for stars awarded, number of questions per game, and time limits per question, allowing flexible game balancing.

User Interaction

Players interact primarily through mouse and keyboard for clicking buttons, selecting options, and text input. The interface supports standard navigation patterns including back buttons, dropdown selectors, and clickable cards. Game-specific interactions include drag-and-drop for puzzles and card-flipping for memory games. Visual feedback is provided through color-coded difficulty levels (green/blue/red), button hover states, timers, and real-time score updates.

3.1.2 Hardware Interfaces

The STARBOOKS Whiz Challenge system interfaces with standard hardware components commonly available in educational and government office settings. All hardware interactions occur through standard operating system APIs and network protocols.

Server PC

- The backend application (Laravel/MongoDB) runs on a standard Windows-based PC serving as the central server
- Minimum specifications: Intel Core i3 or equivalent processor, 4GB RAM, at least 80GB available storage
- Network interface card (NIC) supporting Ethernet connectivity
- The software interacts with the server hardware through standard file system APIs for database storage and network socket APIs for client-server communication

Client PCs (Player Stations)

- Multiple Windows-based PCs serve as player terminals for game access
- Minimum specifications: Intel Core i3 or equivalent processor, 4GB RAM, at least 80GB available storage
- Standard display monitors
- Network interface cards for LAN connectivity
- Standard USB keyboard and mouse for user input
- Web browser support (Chrome, Edge)

Network Equipment

- LAN Router providing wired and/or wireless network connectivity between all PCs
- Standard Ethernet capability (100 Mbps or higher)
- Sufficient ports to accommodate all PCs
- Network cables for wired connections

Data and Control Interactions

- All client-server communication occurs over TCP/IP network protocols through the LAN router
- The software interfaces with network hardware through standard socket programming APIs provided by the operating system
- Database read/write operations interface with local storage through MongoDB's native file system interaction
- User input devices (keyboard, mouse) interface with the application through browser event handling mechanisms
- Display output is rendered through the web browser's graphics rendering engine, which interfaces with the display hardware through standard graphics drivers

3.1.3 Software Interfaces

The STARBOOKS Whiz Challenge system interfaces with various software components to provide full functionality across the client-server architecture.

Operating System Interface

- The system operates on standard Windows operating systems
- Interfaces with the OS through standard file system APIs for local data storage and retrieval
- Utilizes Windows networking stack for TCP/IP communication between server and clients

Web Browser Interface

- The system is accessed through Google Chrome and Microsoft Edge web browsers
- Requires HTML5, CSS3, and JavaScript support for proper rendering and functionality
- Browser provides the runtime environment for the Flutter web application
- Interfaces with browser APIs for user input handling, DOM manipulation, and AJAX communication with the backend server
- Supports WebSocket API for bidirectional real-time communication in Whiz Battle mode

Backend Framework - Laravel

- Laravel PHP framework serves as the server-side application framework
- Provides RESTful API endpoints for client-server communication

- Handles HTTP request/response processing, routing, and middleware services
- Interfaces with the web server (Apache/Nginx via Laragon) for request handling
- Supports WebSocket server integration for real-time battle functionality

Database Management System - MongoDB

- MongoDB serves as the primary NoSQL database for all application data storage
- Laravel interfaces with MongoDB through native MongoDB PHP drivers
- Handles CRUD operations for player profiles, game content, scores, leaderboards, and analytics data
- Provides JSON-based document storage for flexible schema design

Local Development Environment - Laragon

- Laragon provides the local server environment bundling Apache/Nginx, PHP, and MongoDB
- Simplifies server configuration and deployment during development and at event venues
- Manages service startup and port configuration for the local network

Frontend Framework - Flutter Web

- Flutter/Dart framework compiled to web application for cross-browser compatibility
- Generates HTML, CSS, and JavaScript for browser rendering
- Interfaces with Laravel backend through HTTP/HTTPS RESTful API calls and WebSocket connections
- Handles client-side state management and UI rendering

Document Generation and Export

- Excel Export: The system generates Excel spreadsheets (.xlsx format) for exporting analytics data and leaderboard rankings
- PDF Generation: Produces PDF reports for administrative documentation
- Export functionality is accessible through the administrative dashboard with appropriate data formatting and styling

Network Protocol Interface

- All client-server communication occurs over HTTP/HTTPS protocols
- RESTful API architecture for standardized request-response patterns
- JSON format for data serialization and deserialization
- WebSocket protocol for real-time features in Whiz Battle mode, enabling real-time score updates and progress synchronization

Real-Time Communication - WebSocket

- WebSocket connections are established between client browsers and the Laravel backend server for Whiz Battle sessions
- The server acts as a WebSocket hub, routing messages between paired battle participants
- WebSocket connections operate over the local network

Data Exchange Formats

- JSON (JavaScript Object Notation) for API request/response payloads
- XLSX (Excel format) for tabular data exports
- PDF (Portable Document Format) for generated reports

The system does not interface with any external third-party APIs, cloud services, or remote databases. All software components operate within the isolated local network environment to ensure offline functionality.

3.2 Functional Requirements

This section details the specific functional requirements of the STARBOOKS Whiz Challenge system.

3.2.1 F1: Manage Player Account

FR-F1-001: Multi-Step Registration Process. The system shall guide users through a two-step registration process: (1) Privacy Notice acknowledgment, and (2) Account Setup with profile information entry.

FR-F1-002: Player Registration. The system shall allow new users to register by providing the following information: username, password, confirm password, school, age, avatar selection, category, sex, region, province, and city. The system shall validate that all required fields are completed and that the username is unique before creating the account.

FR-F1-003: Player Login. The system shall authenticate players using their username and password credentials. Upon successful authentication, the system shall grant access to the game interface and maintain the user session.

FR-F1-004: Edit Player Profile. The system shall allow authenticated players to edit their profile information including school, age, avatar, category, sex, region, province, and city.

FR-F1-005: Change Password. The system shall allow players to change their password by providing the current password and confirming the new password. The system shall validate that the current password is correct before updating.

FR-F1-006: Player Logout. The system shall provide a logout function that terminates the current player session and returns the user to the login screen.

3.2.2 F2: Play Games

FR-F2-001: Game Mode Selection. The system shall provide access to four game modes from the main menu: Whiz Challenge, Whiz Memory Match, Whiz Puzzle, and Whiz Battle. Each game mode shall be clearly labeled and accessible through dedicated buttons.

FR-F2-002: Category Selection. The system shall allow players to select between categories for all game modes except Whiz Memory Match before starting gameplay.

FR-F2-003: Difficulty Level Selection. The system shall provide three difficulty levels for all game modes: Easy, Average, and Difficult.

Whiz Challenge

FR-F2-004: Question Display. The system shall display questions one at a time with four multiple-choice answer options. Questions shall be randomly selected from the database based on the chosen category and difficulty level.

FR-F2-005: Timer Functionality. The system shall display a countdown timer for each question based on the configured time limit for the selected difficulty level. The timer shall be clearly visible and update in real-time.

FR-F2-006: Answer Selection and Validation. The system shall allow players to select one answer option per question. Upon selection, the system shall immediately validate the answer and provide visual feedback indicating correct or incorrect responses.

FR-F2-007: Score Calculation. The current score shall be displayed and updated in real-time throughout the game.

FR-F2-008: Question Progress Tracking. The system shall display the current question number and total number of questions throughout the gameplay session.

FR-F2-009: Game Completion. Upon answering all questions, the system shall display the overall performance summary.

FR-F2-010: Badge Award for Perfect Score. The system shall automatically award a badge when a player achieves a perfect score in Whiz Challenge for a specific difficulty level and category combination. The badge shall be recorded in the player's badge collection.

FR-F2-011: Pause Functionality. The system shall provide a pause button that temporarily suspends the current game session until the player chooses to resume.

FR-F2-012: Resume Functionality. The system shall provide a resume button that allows the player to continue the game from the exact state where it was paused.

FR-F2-013: Restart Functionality. The system shall provide a restart button that allows the player to restart the game while retaining the previously selected category and difficulty.

FR-F2-014: Exit Game Function. The system shall provide an exit button that allows players to leave the current game session and return to the homepage.

FR-F2-015: Play Again Functionality. The system shall provide a play again button that allows the player to replay the same game and redirects them to the category and difficulty selection screen.

Whiz Memory

FR-F2-016: Memory Card Grid Display. The system shall display a grid of face-down cards containing matching pairs of images related to the selected category and difficulty level.

FR-F2-017: Card Flip Mechanism. The system shall allow players to click on cards to flip them face-up, revealing the image underneath. Players shall be able to flip two cards per turn.

FR-F2-018: Matching Logic. The system shall compare the two flipped cards. If they match, the cards shall remain face-up. If they do not match, the cards shall flip back face-down after a brief display period.

FR-F2-019: Move Counter. The system shall track and display the total number of moves (card flip attempts) made by the player throughout the game session.

FR-F2-020: Memory Game Timer. The system shall display an elapsed time counter showing how long the player has been playing the current memory game session.

FR-F2-021: Memory Game Completion. The system shall detect when all pairs have been matched and display a completion message with final time and total moves.

FR-F2-022: Star Scoring. The system shall award stars on players based on their performance.

FR-F2-023: Pause Functionality. The system shall provide a pause button that temporarily suspends the current game session until the player chooses to resume.

FR-F2-024: Resume Functionality. The system shall provide a resume button that allows the player to continue the game from the exact state where it was paused.

FR-F2-025: Restart Functionality. The system shall provide a restart button that allows the player to restart the game while retaining the previously selected difficulty.

FR-F2-026: Exit Game Function. The system shall provide an exit button that allows players to leave the current game session and return to the homepage.

FR-F2-027: Play Again Functionality. The system shall provide a play again button that allows the player to replay the same game and redirects them to the difficulty selection screen.

Whiz Puzzle

FR-F2-028: Puzzle Image Selection. The system shall select an image from the database based on the chosen category and difficulty level to serve as the puzzle.

FR-F2-029: Puzzle Piece Generation. The system shall divide the selected image into jigsaw puzzle pieces. The number of pieces shall vary based on the difficulty level (fewer pieces for Easy, more pieces for Difficult).

FR-F2-030: Drag-and-Drop Interface. The system shall allow players to drag puzzle pieces using the mouse and drop them into position on the puzzle board.

FR-F2-031: Piece Snapping Mechanism. The system shall automatically snap puzzle pieces into the correct position when placed within a proximity threshold, providing visual and functional feedback.

FR-F2-032: Star Scoring. The system shall award stars on players based on their performance.

FR-F2-033: Puzzle Timer. The system shall display an elapsed time counter showing how long the player has been working on the current puzzle.

FR-F2-034: Move Counter. The system shall track and display the total number of moves made by the player throughout the game session.

FR-F2-035: Puzzle Completion Detection. The system shall detect when all puzzle pieces are correctly positioned and display a completion message with the final time and congratulatory feedback.

FR-F2-036: Pause Functionality. The system shall provide a pause button that temporarily suspends the current game session until the player chooses to resume.

FR-F2-037: Resume Functionality. The system shall provide a resume button that allows the player to continue the game from the exact state where it was paused.

FR-F2-038: Restart Functionality. The system shall provide a restart button that allows the player to restart the game while retaining the previously selected category and difficulty.

FR-F2-039: Exit Game Function. The system shall provide an exit button that allows players to leave the current game session and return to the homepage.

FR-F2-040: Play Again Functionality. The system shall provide a play again button that allows the player to replay the same game and redirects them to the category and difficulty selection screen.

Whiz Battle (1v1 Quiz)

FR-F2-041: Game Code Generation. The system shall generate a unique game code when a player initiates a Whiz Battle session. This code shall be displayed to both players and used for matchmaking.

FR-F2-042: Player Matchmaking. The system shall allow two players to join the same battle session by entering or sharing the game code. The system shall display both players' avatars and usernames in a versus format.

FR-F2-043: Battle Start Synchronization. The system shall ensure both players start the quiz simultaneously when the host player clicks the "START" button. Both players shall receive the same set of questions in the same order.

FR-F2-044: Real-Time Battle Progress. The system shall display both players' progress and scores in real-time during the battle, allowing each player to see their opponent's performance.

FR-F2-045: Battle Scoring. The system shall award stars for correct answers following the same rules as Whiz Challenge. The system shall track individual scores for both players throughout the battle.

FR-F2-046: Battle Completion and Winner Determination. Upon completion of all questions, the system shall compare final scores and declare the winner. Results shall be displayed to both players showing final scores and performance statistics.

FR-F2-047: Badge Award for Winner. The system shall automatically award a badge when a player wins a battle for a specific difficulty level and category combination. The badge shall be recorded in the player's badge collection.

3.2.3 F3: Reward Redemption

FR-F3-001: Badge Accumulation Tracking. The system shall track the number of badges earned by each player for each difficulty level (Easy, Average, Difficult) across all categories in Whiz Challenge and Whiz Battle.

FR-F3-002: Reward Eligibility Check. The system shall determine that a player is eligible to claim a reward when they have accumulated exactly three (3) badges within a single difficulty level.

FR-F3-003: Reward Claim Request. The system shall allow eligible players to submit a reward claim request through the player interface, specifying the difficulty level for which they wish to claim a reward.

FR-F3-004: Reward Claim Logging. Upon submission of a reward claim request, the system shall log the claim with timestamp, player username, difficulty level, and claim status (pending) in the administrative dashboard.

FR-F3-005: Admin Reward Verification Interface. The system shall provide administrators with access to the list of pending reward claims in the administrative dashboard, displaying player usernames and claimed difficulty levels.

FR-F3-006: Badge Collection Lookup. The system shall allow administrators to look up a specific player's badge collection by username, displaying all earned badges organized by difficulty level.

FR-F3-007: Admin Reward Grant Function. The system shall provide administrators with the ability to grant reward claims after verifying that the player has three badges in the specified difficulty level.

FR-F3-008: Badge Reset Upon Reward Grant. When an administrator grants a reward, the system shall automatically reset the player's badge count to zero (0) for that specific difficulty level, removing the three redeemed badges from the player's collection.

FR-F3-009: Reward Grant Confirmation. The system shall update the reward claim status to "granted" with timestamp and administrator username, maintaining a permanent record of the transaction.

3.2.4 F4: Manage Admin Account

FR-F4-001: Admin Login. The system shall provide a separate login interface for administrators accessible through the "ADMIN" link. Administrators shall authenticate using admin-specific credentials. The system shall validate admin credentials against the admin user database and grant access to the administrative dashboard upon successful authentication.

FR-F4-002: Add New Admin. The system shall allow authorized administrators to create new admin accounts by providing username and password. The system shall validate that new admin usernames are unique across the admin database before creating the account.

FR-F4-003: Edit Admin Profile. The system shall allow administrators to edit existing admin profile information.

FR-F4-004: Admin Password Change. The system shall allow administrators to change their own password or reset passwords for other admin accounts with appropriate authorization.

FR-F4-005: Delete Admin Account. The system shall allow authorized administrators to delete admin accounts. The system shall require confirmation before permanent deletion and maintain audit logs of deleted accounts.

FR-F4-006: Admin Logout. The system shall provide a logout function that terminates the admin session and returns the user to the admin login screen.

3.2.5 F5: Manage Quiz Settings

Question Management

FR-F5-001: Question List Display. The system shall display a searchable and filterable list of all questions in the database, showing question text preview, category, difficulty level, and other information.

FR-F5-002: Question Search and Filter. The system shall provide search functionality to find questions by keyword and filter options by category, difficulty level, and topic tags for efficient content management.

FR-F5-003: Add New Question. The system shall allow administrators to add new quiz questions by providing question text, four answer options, correct answer designation, category (Science or Math), and difficulty level (Easy, Average, Difficult). The system shall validate that question text and all answer options are provided and contain sufficient content before saving a new question to the database.

FR-F5-004: Edit Question Information. The system shall allow administrators to edit existing questions, modifying question text, answer options, correct answer, category, and difficulty level.

FR-F5-005: Delete Question. The system shall allow administrators to delete questions from the database. The system shall require confirmation before permanent deletion to prevent accidental data loss.

Game Configuration

FR-F5-006: Set Number of Questions. The system shall allow administrators to configure the number of questions per game session for each difficulty level (Easy, Average, Difficult). This setting shall apply to Whiz Challenge and Whiz Battle game modes.

FR-F5-007: Set Number of Stars. The system shall allow administrators to configure the number of stars awarded per correct answer for each difficulty level.

FR-F5-008: Set Timer Duration. The system shall allow administrators to configure the time limit per question (in seconds) for each difficulty level. The timer setting shall apply to Whiz Challenge and Whiz Battle modes.

3.2.6 F6: Manage Player List

FR-F6-001: View Player List. The system shall display a comprehensive list of all registered players showing username, avatar, registration date, total rewards, and demographic information.

FR-F6-002: Add New Player. The system shall allow administrators to manually create new player accounts by providing all required registration information (username, password, school, age, avatar, category, sex, region, province, city).

FR-F6-003: Edit Player Profile. The system shall allow administrators to modify player profile information including school, age, avatar, category, sex, region, province, and city.

FR-F6-004: Player Password Reset. The system shall allow administrators to reset player passwords for account recovery purposes.

FR-F6-005: View Player Badge Collection. The system shall allow administrators to view detailed badge collections for individual players, showing badges earned by difficulty level.

FR-F6-006: Delete Player Account. The system shall allow administrators to delete player accounts. The system shall require confirmation before permanent deletion.

3.2.7 F7: View Analytics Dashboard

FR-F7-001: Analytics Dashboard Access. The system shall provide administrators with access to a comprehensive analytics dashboard displaying key performance indicators and visualization panels.

FR-F7-002: Total Registered Players. The system shall display the total count of registered players in the database as a prominent summary metric.

FR-F7-003: Most Played Game Mode. The system shall analyze gameplay data and display which game mode (Whiz Challenge, Whiz Memory, Whiz Puzzle, or Whiz Battle) has been played most frequently.

FR-F7-004: Average Session Duration. The system shall calculate and display the average time players spend in game sessions across all game modes, presented in minutes and seconds format.

FR-F7-005: Male vs Female Registered Players. The system shall generate a visualization (chart or graph) showing the gender distribution of registered players, comparing male and female player counts.

FR-F7-006: Age Distribution. The system shall generate a visualization showing the distribution of players across different age ranges, helping identify primary demographic segments.

FR-F7-007: Players by Region. The system shall generate a visualization displaying the number of registered players from each region, sorted by player count to identify regions with highest engagement.

FR-F7-008: Male vs Female Players per Game Mode. The system shall generate a comparative visualization showing gender distribution of players for each game mode, revealing game mode preferences by gender.

FR-F7-009: Rewards Distribution by Gender and Level. The system shall generate a visualization showing how rewards are distributed across genders and difficulty levels (Easy, Average, Difficult), providing insights into achievement patterns.

FR-F7-010: Most Played Game Mode by Age. The system shall generate a visualization showing game mode preferences segmented by age groups, identifying which age demographics prefer which game modes.

FR-F7-011: Data Export Functionality. The system shall provide an "Export" button allowing administrators to download analytics data for reporting.

FR-F7-012: Data Filtering Options. The system shall provide filtering capabilities on analytics panels, allowing administrators to sort, filter, and customize data views using filter icons.

FR-F7-013: Real-Time Data Updates. The system shall provide a refresh mechanism to update analytics data with the most recent information from the database.

3.2.8 F8: View Leaderboard

FR-F8-001: Leaderboard Access. The system shall provide both players and administrators access to view leaderboards through dedicated navigation options labeled "Leaderboard."

FR-F8-002: Leaderboard Ranking. The system shall rank players based on the total number of rewards claimed. Players with more claimed rewards shall appear higher in the ranking.

FR-F8-003: Top 20 Display Limit. The system shall display only the top 20 ranked players to maintain focus on top performers and optimize display performance.

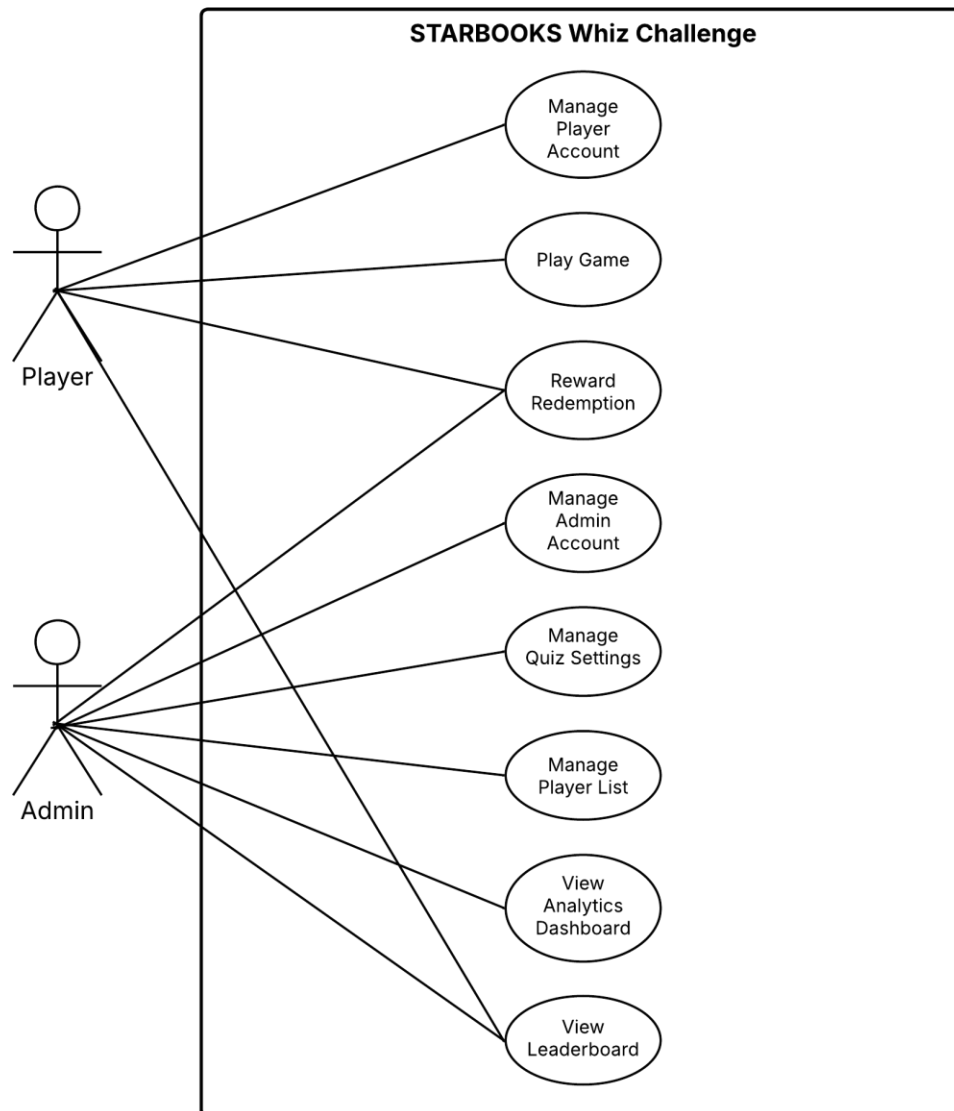
FR-F8-004: Player Information Display. For each ranked player, the system shall display rank position, player avatar thumbnail, username, total rewards, and reward breakdown by difficulty level (Easy, Average, Difficult). The system shall also display the date and time of each player's most recent reward claim in the format "MM/DD/YYYY HH:MM" to show leaderboard recency.

FR-F8-005: Real-Time Leaderboard Updates. The system shall update leaderboard rankings automatically as players complete games and claim rewards, ensuring current and accurate competitive standings.

FR-F8-006: Admin Export Functionality. The system shall provide an "Export" button visible only to administrators, allowing them to download complete leaderboard data for record-keeping and reporting purposes.

FR-F8-007: Tie-Breaking Rules. When multiple players have the same number of rewards or stars, the system shall use the most recent claim timestamp as a tie-breaker, with more recent activity ranking higher.

3.3 Use Case Model



The Use Case Diagram above provides a high-level view of how players and admins interact with the STARBOOKS Whiz Challenge system. These diagrams map out the different actors and the specific functions they can perform, making it easier to understand the system's overall scope and behavior.

For the complete set of fully dressed use cases, see the following [link](#).

4 Other Non-functional Requirements

4.1 Performance Requirements

The following performance requirements ensure the STARBOOKS Whiz Challenge system operates efficiently during DOST events with multiple concurrent users in an offline LAN environment.

4.1.1 Response Time Requirements

PR-001: Page Load Time. The system shall load any page or screen within seconds under normal operating conditions on the local network.

PR-002: User Interface Interaction Response. The system shall respond to user interactions (button clicks, dropdown selections, form submissions) within seconds, providing immediate visual feedback to confirm user actions.

PR-003: API Response Time. The system shall process and respond to API requests between the client and server within seconds for standard operations under normal load conditions.

PR-004: Login Authentication. The system shall complete user authentication (player or admin login) and redirect to the appropriate dashboard within seconds of credential submission.

4.1.2 Game Performance Requirements

PR-005: Question Loading. The system shall load and display the next question in Whiz Challenge or Whiz Battle mode within second after the player submits an answer or the timer expires.

PR-006: Timer Accuracy. The timer for all game modes shall maintain accuracy over the duration of a complete game session to ensure fair gameplay.

PR-007: Answer Validation Feedback. The system shall validate answer submissions and provide visual feedback (correct/incorrect indication) within seconds of player selection to maintain game flow.

PR-008: Real-Time Battle Updates. In Whiz Battle mode, the system shall synchronize and display opponent progress updates to provide near real-time competitive experience.

PR-009: Memory Card Flip Animation. The system shall complete card flip animations in Whiz Memory game within seconds to maintain smooth gameplay without frustrating delays.

PR-010: Puzzle Piece Drag Response. The system shall provide smooth drag-and-drop functionality for puzzle pieces with response time under seconds for cursor tracking to ensure intuitive manipulation.

4.1.3 Concurrent User Performance

PR-011: Simultaneous Player Support. The system shall support a minimum of 10 concurrent active players playing games simultaneously without performance degradation, measured by maintaining response times.

PR-012: Scalability to Peak Load. The system shall support up to 15 concurrent users (players and administrators) during peak event periods while maintaining response times within acceptable ranges.

PR-013: Concurrent Game Sessions. The system shall support at least 10 simultaneous active game sessions across all game modes without server resource exhaustion or game interruption.

PR-014: Database Transaction Throughput. The system shall process a minimum of 100 database transactions per minute to support multiple concurrent users saving progress, updating scores, and retrieving content.

4.1.4 Leaderboard and Analytics Performance

PR-015: Leaderboard Update Frequency. The system shall update leaderboard rankings within seconds after a player completes a game or claims a reward to maintain competitive engagement.

PR-016: Leaderboard Load Time. The system shall load and display the complete top 20 leaderboard within seconds when accessed by players or administrators.

PR-017: Analytics Dashboard Load Time. The system shall load the complete analytics dashboard with all visualization panels within seconds when accessed by administrators.

PR-018: Analytics Data Refresh. The system shall refresh analytics data and regenerate visualizations within seconds when the administrator clicks the refresh button.

PR-019: Report Generation Time. The system shall generate and prepare Excel or PDF exports within seconds for datasets containing records.

4.1.5 Data Management Performance

PR-020: Database Query Response. The system shall execute standard database queries with an average response time around milliseconds.

PR-021: Complex Query Performance. The system shall execute complex analytical queries within seconds to support dashboard functionality.

PR-022: Data Write Operations. The system shall complete data write operations (save game results, update player profile, add question) within second to ensure data persistence without noticeable delay.

PR-023: Batch Operations. The system shall process batch operations within seconds.

4.1.6 Capacity Requirements

PR-024: Question Database Capacity. The system shall support storage and efficient retrieval of at least 100 questions across all categories and difficulty levels without performance degradation.

PR-025: Player Account Capacity. The system shall support a minimum of 100 registered player accounts with complete profile information, game history, and badge collections while maintaining query performance standards.

PR-026: Game History Storage. The system shall store and retrieve complete game session history for at least 100 game sessions per player account without impacting system performance.

PR-027: Concurrent Admin Operations. The system shall support up to 5 administrators accessing the dashboard simultaneously and performing management operations without conflicts or performance issues.

4.1.7 Network Performance Requirements

PR-028: LAN Network Latency. The system shall operate efficiently with network latency around milliseconds within the local area network to accommodate standard LAN router performance.

PR-029: Connection Stability. The system shall handle temporary network interruptions gracefully by queuing operations and resuming when connection is restored, without data loss for completed game actions.

4.1.8 System Resource Requirements

PR-030: Server Memory Usage. The server application shall operate within 4GB of RAM under normal conditions to ensure compatibility with standard PC specifications.

PR-031: Client Browser Performance. The client web application shall operate smoothly on standard web browsers with at least 4GB of available system RAM and shall not cause browser unresponsiveness or crashes during extended gameplay sessions.

PR-032: Database Storage Efficiency. The MongoDB database shall maintain efficient storage utilization, with database size growing no more than 100MB per 1,000 player accounts including game history and analytics data.

4.2 Safety and Security Requirements

The following safety and security requirements ensure the STARBOOKS Whiz Challenge system protects user data, prevents unauthorized access, and complies with applicable data privacy regulations.

4.2.1 Data Privacy and Protection

SS-001: Data Privacy Act Compliance. The system shall comply with all provisions of the Philippine Data Privacy Act of 2012 (Republic Act No. 10173) regarding the collection, storage, processing, and protection of personal information from players, particularly minors under 18 years of age.

SS-002: Informed Consent. The system shall display a Privacy Notice during the registration process informing users about what data is collected, how it will be used, and that it will be stored locally for DOST-STII educational purposes. Users must acknowledge the privacy notice before proceeding with account creation.

SS-003: Minimal Data Collection. The system shall collect only the minimum personal information necessary for system functionality and analytics purposes: username, age, gender, school, region, province, and city. No sensitive personal information such as full names, addresses, contact numbers, or email addresses shall be collected from players.

SS-004: Data Confidentiality. All player personal information and game performance data shall be treated as confidential and accessible only to authorized DOST-STII administrators through the secure administrative dashboard.

4.2.2 Authentication and Access Control

SS-005: User Authentication Requirement. The system shall require all users (players and administrators) to authenticate with valid credentials (username and password) before accessing any system functionality.

SS-006: Password Storage Security. The system shall store all passwords using strong one-way cryptographic hashing algorithms with unique salts per user. Passwords shall never be stored in plain text or reversible encrypted format.

SS-007: Password Complexity Requirements. The system shall enforce minimum password requirements: at least 8 characters in length, containing a mix of uppercase and lowercase letters, numbers, and special characters to prevent weak passwords.

SS-008: Role-Based Access Control. The system shall implement role-based access control distinguishing between player accounts and administrator accounts. Players shall only access game functionality and their own profiles, while administrators shall have exclusive access to management and analytics functions.

SS-009: Admin Authentication Separation. The system shall maintain separate authentication pathways for administrators through a dedicated admin login interface, preventing players from accessing administrative functions even if credentials are compromised.

4.2.3 Network Security

SS-010: Network Isolation. The system shall operate on an isolated local area network without internet connectivity during events, preventing external network attacks and unauthorized remote access.

SS-011: Network Access Control. The LAN router shall be configured to prevent unauthorized devices from connecting to the network through MAC address filtering or other access control mechanisms as determined by DOST-STII technical staff.

SS-012: Port Security. The server shall only expose necessary network ports (HTTP/HTTPS) for client communication, with all other ports closed or filtered to minimize attack surface.

SS-013: Protection Against Network Attacks. The system shall implement basic protections against common network attacks including request rate limiting to prevent denial-of-service attempts from malicious clients on the LAN.

4.2.4 Safety Requirements for Young Users

SS-014: Age-Appropriate Content. All game content shall be reviewed and approved by DOST-STII content specialists to ensure age-appropriateness and educational value for elementary through senior high school students.

SS-015: Protection of Minors' Data. Given that the primary users are minors (students under 18), the system shall implement enhanced data protection measures and shall not share, sell, or transfer player data to any third parties under any circumstances.

SS-016: Safe Gaming Environment. The system shall provide a safe, controlled gaming environment free from external communication features that could expose young users to inappropriate content or online predators.

4.2.5 Operational Safety

SS-017: System Stability. The system shall be designed to prevent crashes, data corruption, or unexpected behavior that could disrupt DOST events or result in loss of player progress during active game sessions.

SS-018: Graceful Error Recovery. In the event of errors or exceptions, the system shall recover gracefully without exposing sensitive information, corrupting data, or requiring complete system restart when possible.

SS-019: Data Backup Procedures. DOST-STII administrators shall implement regular backup procedures for the database to prevent data loss due to hardware failure, software errors, or accidental deletion. Backups shall be performed at least daily during multi-day events.

4.3 Software Quality Attributes

The following quality attributes define non-functional characteristics critical to the success and long-term viability of the STARBOOKS Whiz Challenge system.

4.3.1 Usability

Usability is vital for the STARBOOKS Whiz Challenge system given its target audience of elementary through senior high school students with varying levels of computer literacy and technical experience.

SQA-U-001: Intuitive Navigation. The system shall provide intuitive navigation with clearly labeled buttons, consistent placement of navigation elements across all screens, and visual hierarchy that guides users naturally through workflows. New players shall be able to complete registration and start their first game within 5 minutes without external assistance.

SQA-U-002: Visual Clarity. The system shall use age-appropriate visual design with high contrast between text and backgrounds, legible font sizes, and color-coded elements that are distinguishable for users with common color vision deficiencies.

SQA-U-003: Minimal Learning Curve. The player interface shall be designed for immediate comprehension, requiring no user manual or training for basic gameplay. Game rules and controls shall be self-evident through visual cues, icons with text labels, and consistent interaction patterns across all game modes.

SQA-U-004: Error Prevention and Recovery. The system shall prevent user errors through input validation, confirmation dialogs for destructive actions, and clear indication of required fields. When errors occur, the system shall provide specific, actionable error messages explaining what went wrong and how to correct it.

SQA-U-005: Responsive Feedback. The system shall provide immediate visual feedback for all user interactions including button hover states, click animations, loading indicators for operations taking longer than 1 second, and progress indicators for multi-step processes.

SQA-U-006: Administrative Efficiency. The administrative dashboard shall enable DOST-STII staff to complete common tasks efficiently. The interface shall minimize clicks required for frequent operations.

4.3.2 Maintainability

Maintainability ensures DOST-STII technical staff can effectively manage, update, and troubleshoot the system after deployment without requiring the original development team.

SQA-M-001: Code Documentation. All source code shall include comprehensive inline comments explaining complex logic, business rules, and non-obvious implementation decisions.

SQA-M-002: Modular Architecture. The system shall be designed with modular architecture separating concerns into distinct components: presentation layer (Flutter UI), business logic layer (Laravel controllers and services), and data access layer (MongoDB repositories). Changes to one module shall not require modifications to unrelated modules, enabling independent updates and bug fixes.

SQA-M-003: Database Schema Documentation. Complete database schema documentation shall be provided including entity-relationship diagrams, field descriptions, data types, constraints, indexes, and relationships between collections. This enables DOST-STII staff to understand data structures for reporting, backup, and potential future enhancements.

SQA-M-004: Version Control and Change Tracking. All source code shall be maintained in a version control system (Git) with meaningful commit messages documenting what changed and why.

SQA-M-005: Standardized Coding Conventions. The codebase shall adhere to established coding standards for each technology ensuring consistent formatting, naming conventions, and code organization. This reduces cognitive load for future maintainers and simplifies code reviews.

SQA-M-006: Dependency Management. All external libraries and dependencies shall be explicitly declared with version specifications in dependency management files.

4.3.3 Reliability

Reliability ensures the system operates consistently and correctly during DOST events, maintaining data integrity and avoiding disruptions that could negatively impact the educational experience.

SQA-R-001: Consistent Game State. The system shall maintain consistent game state throughout player sessions. If a player's browser crashes or connection is lost during gameplay, the system shall either preserve the game state for resumption.

SQA-R-002: Backup and Recovery. The system shall support database backup and restoration procedures enabling DOST-STII staff to create manual backups before events and restore from backups in case of data corruption or hardware failure. Backup procedures shall be documented and tested to ensure successful recovery.

4.3.4 Portability

Portability ensures the system can be deployed across different hardware configurations and Windows environments that DOST-STII may encounter at various event venues.

SQA-P-001: Cross-Browser Compatibility. The system shall function correctly across multiple modern web browsers without requiring browser-specific code or special configurations. Visual rendering and functionality shall be consistent across supported browsers.

SQA-P-002: Hardware Independence. The system shall operate on standard Windows PCs meeting minimum specifications (Intel Core i3 or equivalent, 4GB RAM) without requiring specific hardware brands, models, or proprietary components.

SQA-P-003: Windows Version Compatibility. The system shall be compatible with Windows 10 and Windows 11 operating systems, utilizing only APIs and system calls available across these versions. Installation and configuration procedures shall be documented for both OS versions.

SQA-P-004: Self-Contained Deployment. The system shall be packaged with all dependencies through Laragon, enabling deployment to new event venues by copying the application folder and starting services, without requiring complex installation procedures, internet downloads, or system-wide installations that may require administrator privileges on venue PCs.

4.3.5 Testability

Testability ensures the system can be effectively validated for correctness, enabling quality assurance processes and confidence in system behavior.

SQA-T-001: Unit Test Coverage. Critical business logic components shall have unit test coverage exceeding 80%, with tests validating both expected functionality and edge cases. Test results shall be documented for DOST-STII quality assurance records.

SQA-T-002: Test Data Management. The system shall support loading test datasets with known configurations (predefined questions, sample players, controlled scores) enabling consistent, repeatable testing scenarios for validation before events. Test data shall be clearly distinguished from production data.

4.3.6 Performance Efficiency

Performance efficiency ensures optimal resource utilization given the offline, LAN-based deployment constraints and standard PC hardware specifications.

SQA-PE-001: Scalable Architecture. While Version 1.0 targets minimum of 10 concurrent users, the system architecture shall avoid hardcoded limits and use scalable patterns (connection pooling, stateless services, efficient data structures) that could support growth to 10+ users with hardware upgrades rather than architectural redesign.

SQA-PE-002: Memory Management. The system shall implement proper memory management practices including closing database connections after use, clearing cached data when no longer needed, and avoiding memory leaks from event listeners or circular references that could degrade performance during extended events.

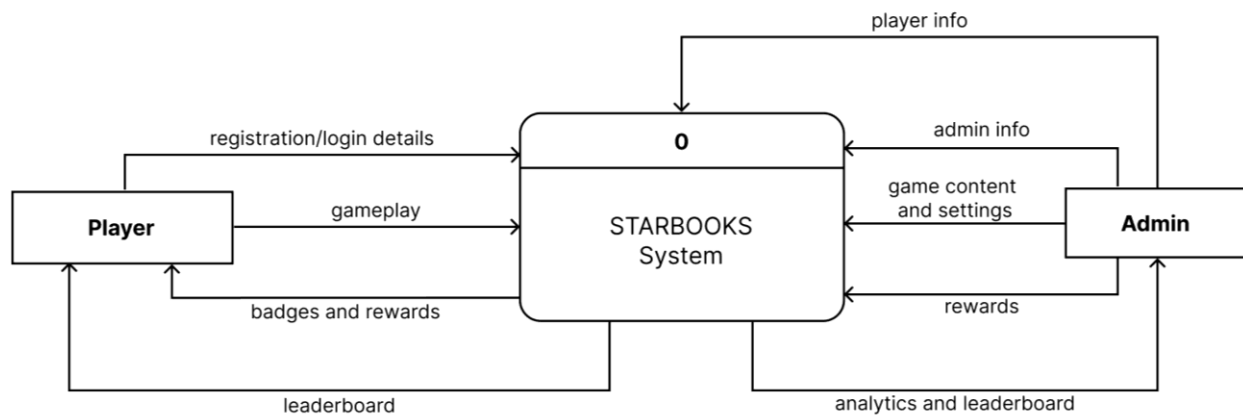
SQA-PE-003: Network Efficiency. Client-server communication shall be optimized through data compression and incremental loading techniques to minimize network traffic and improve responsiveness on standard LAN infrastructure.

5 Other Requirements

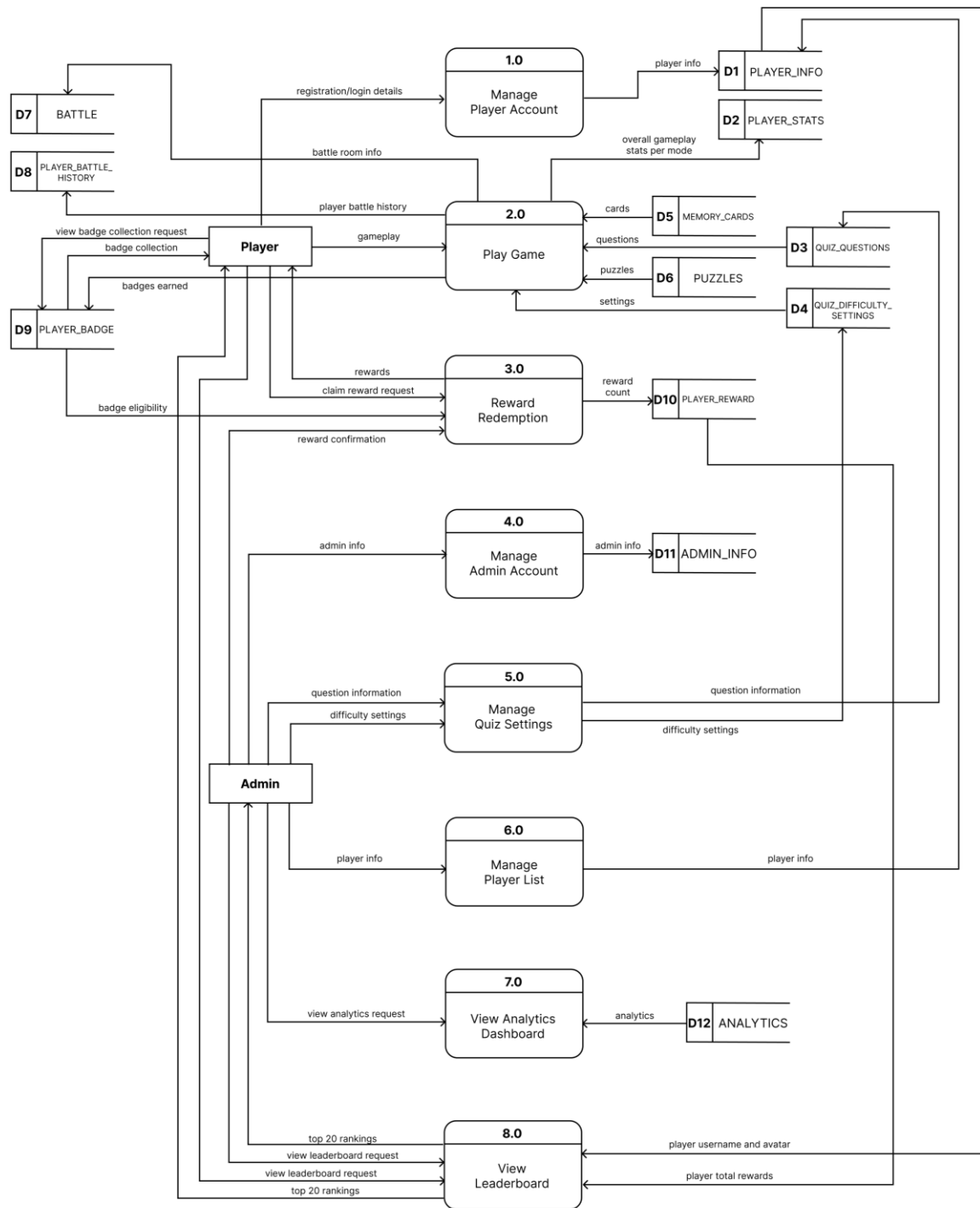
5.1 Dataflow Diagrams

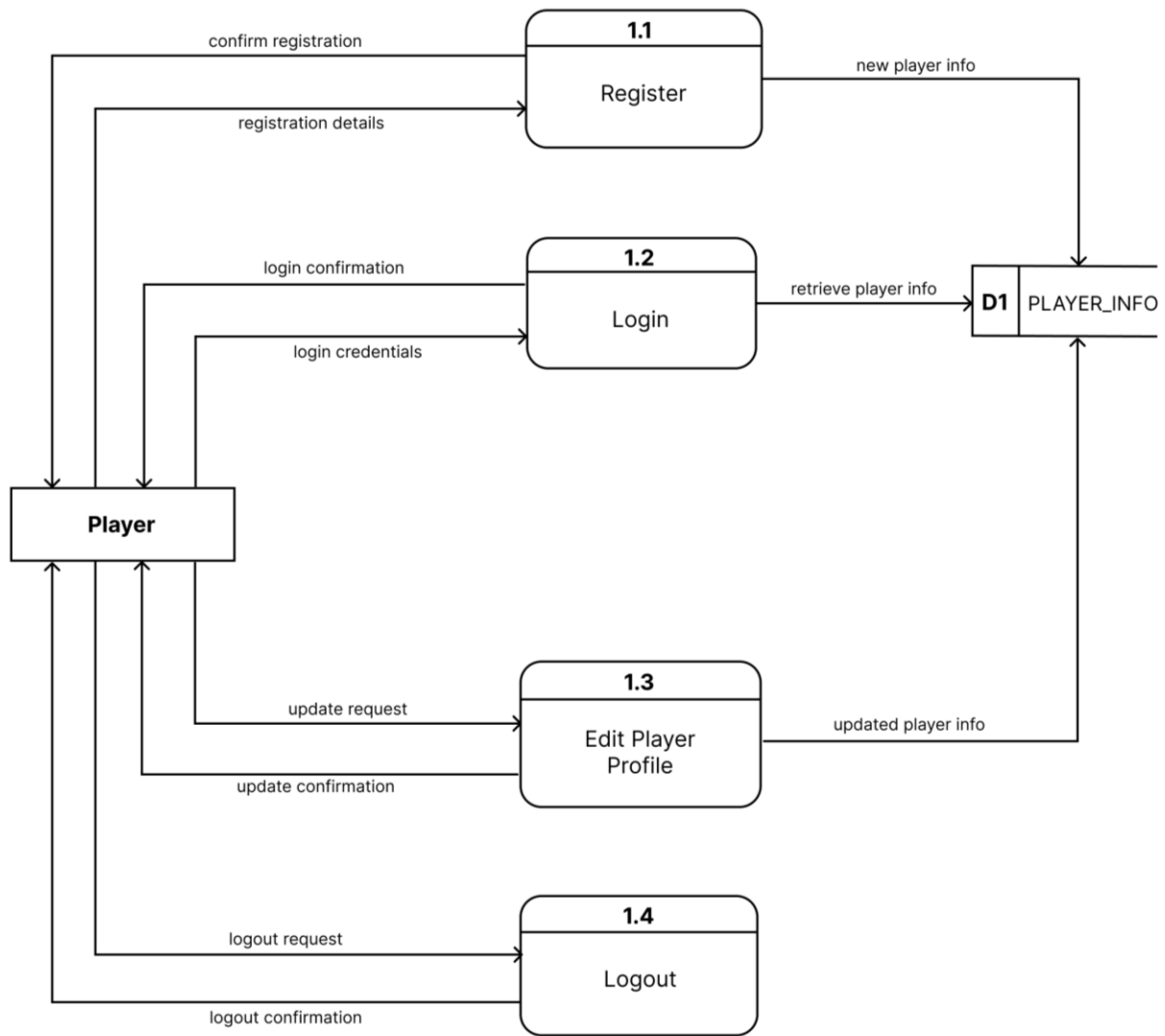
The following Data Flow Diagrams (DFDs) show how information moves through the STARBOOKS Whiz Challenge system. The Level 0 presents the system as a single, high-level process and shows how it interacts with external entities like players and admins. The Level 1 breaks this main process into its major sub-processes, giving a clearer view of how the system handles key functions. Finally, the Level 2 provide a deeper look into specific processes, showing the detailed steps and data stores involved.

Level 0

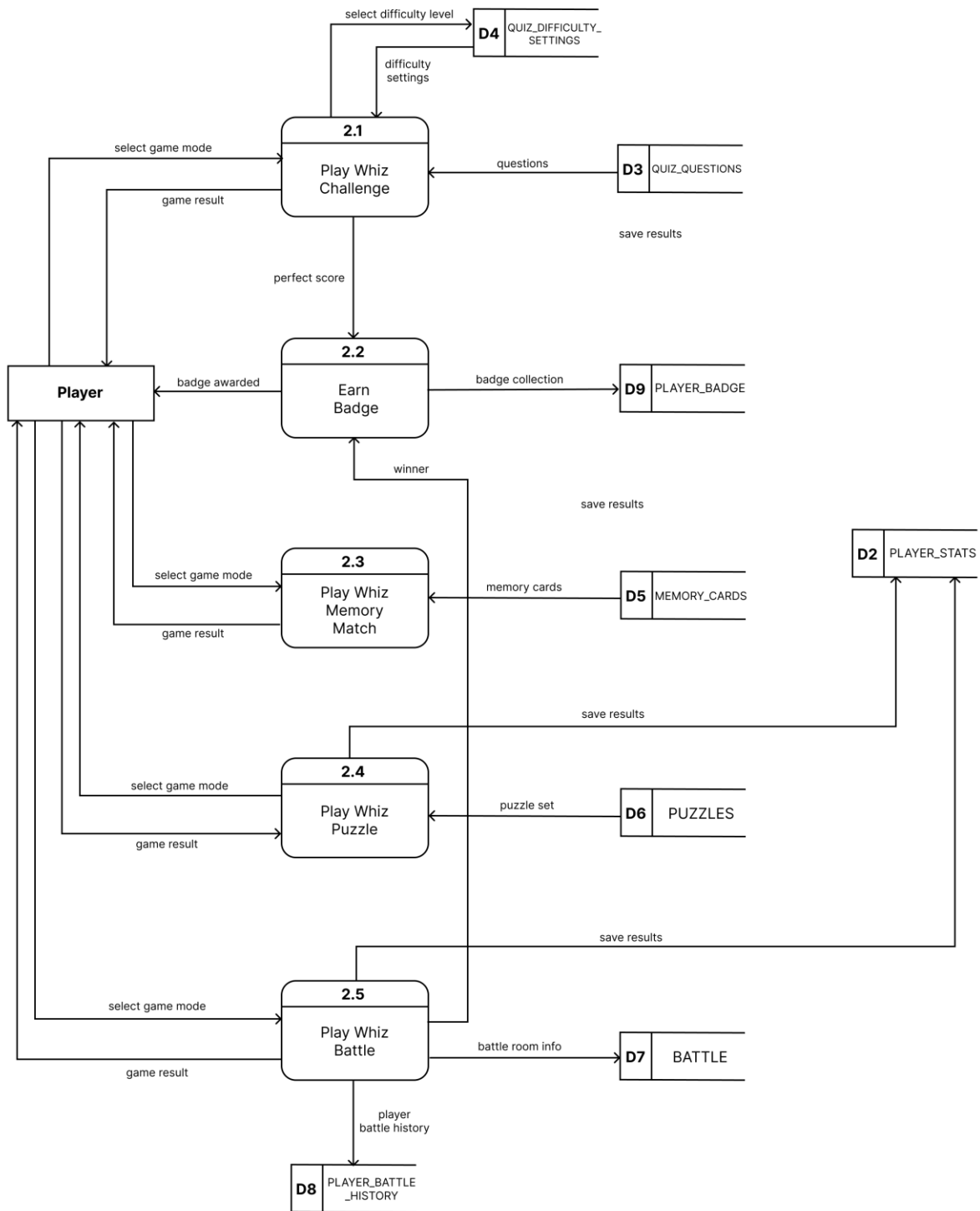


Level 1

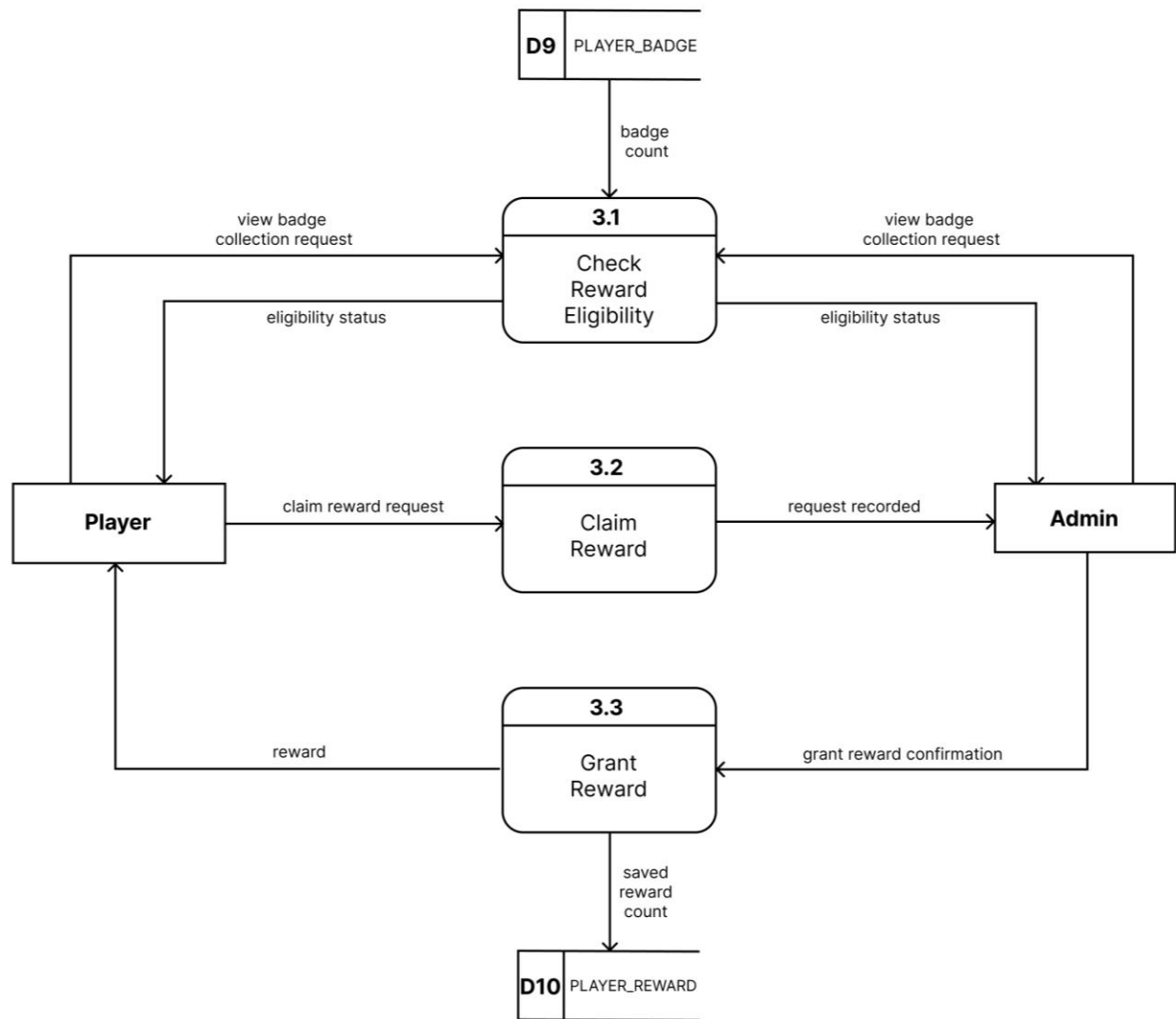


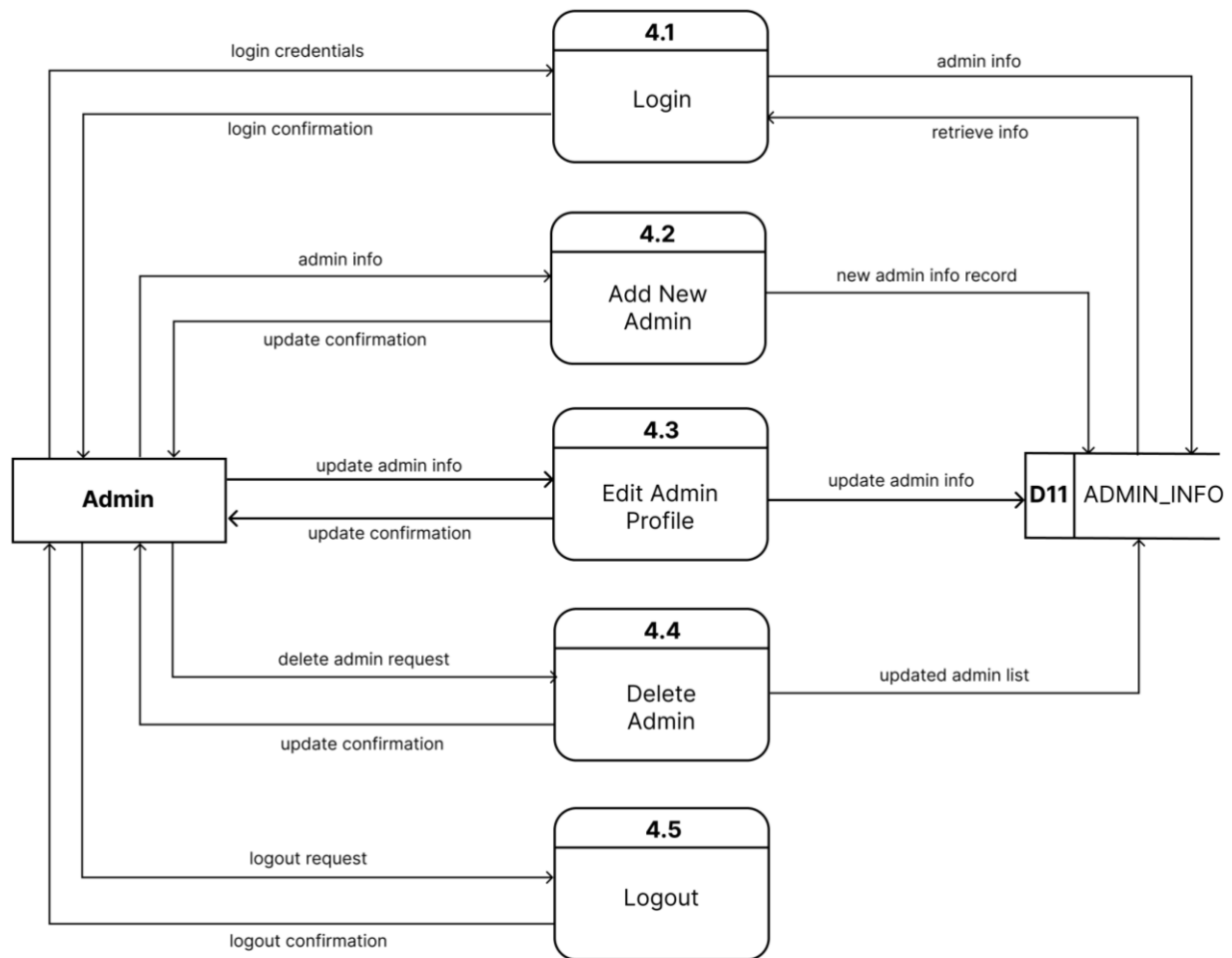
Level 2: 1.0 Manage Player Account

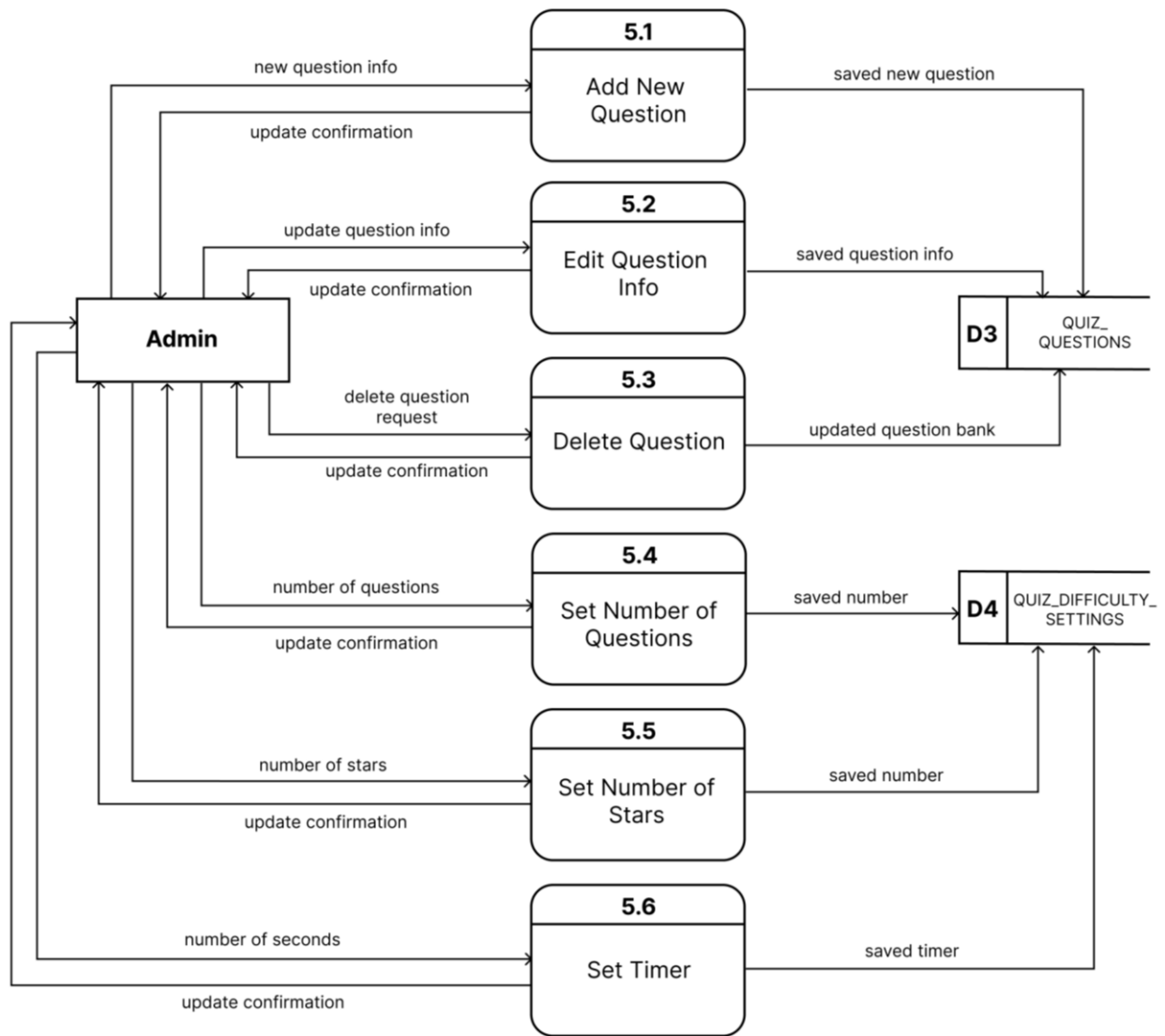
Level 2: 2.0 Play Game

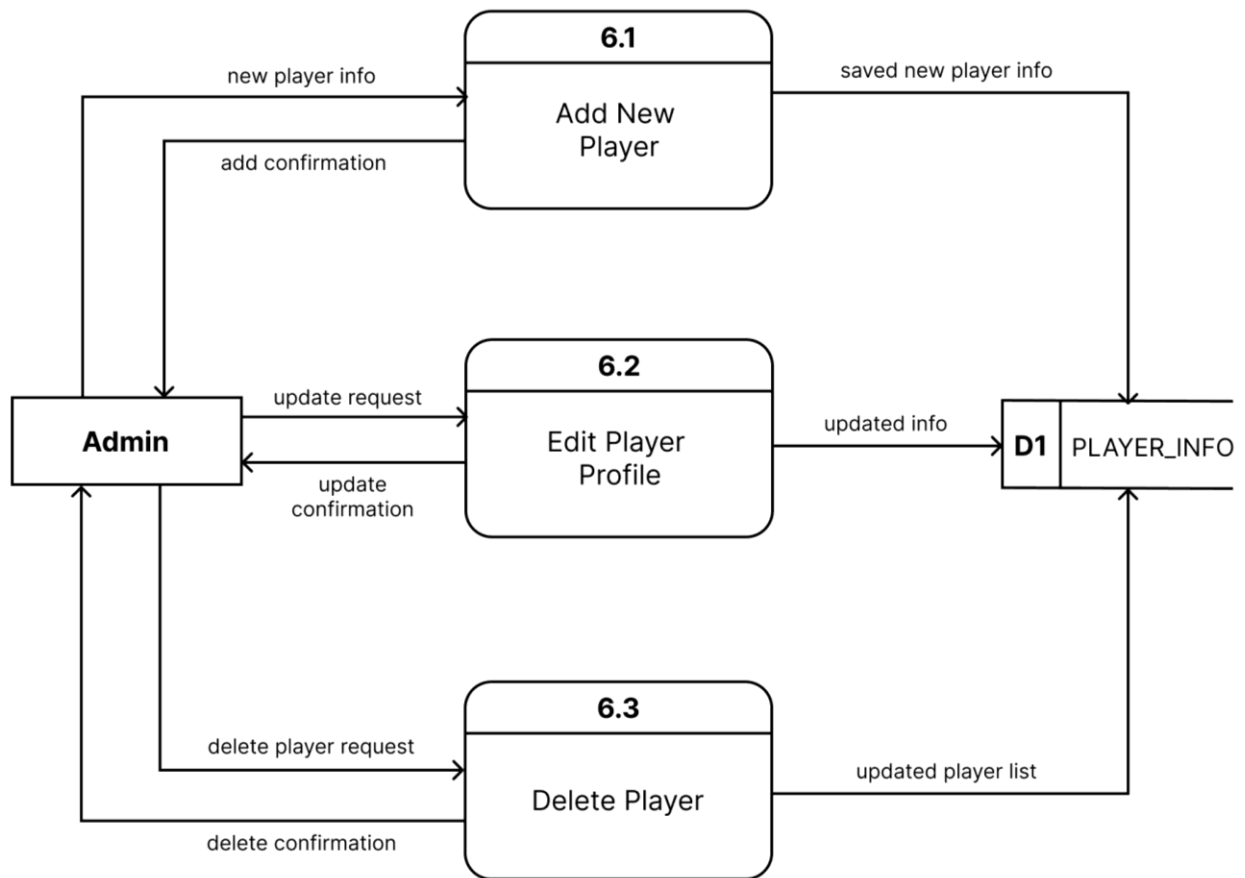


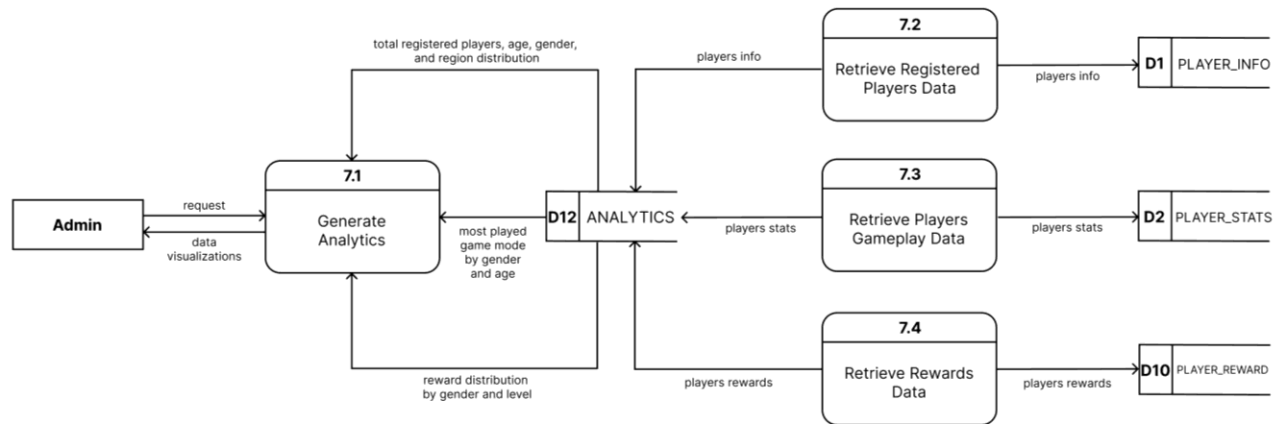
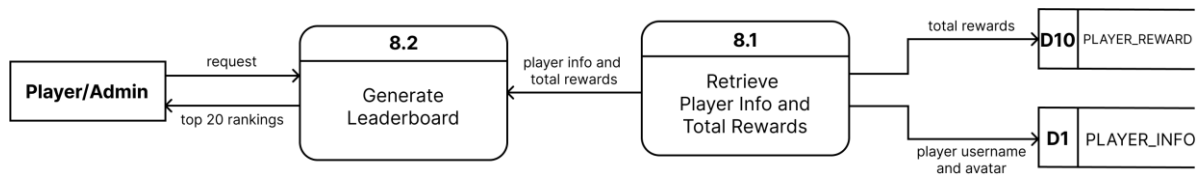
Level 2: 3.0 Reward Redemption



Level 2: 4.0 Manage Admin Account

Level 2: 5.0 Manage Quiz Settings

Level 2: 6.0 Manage Player List

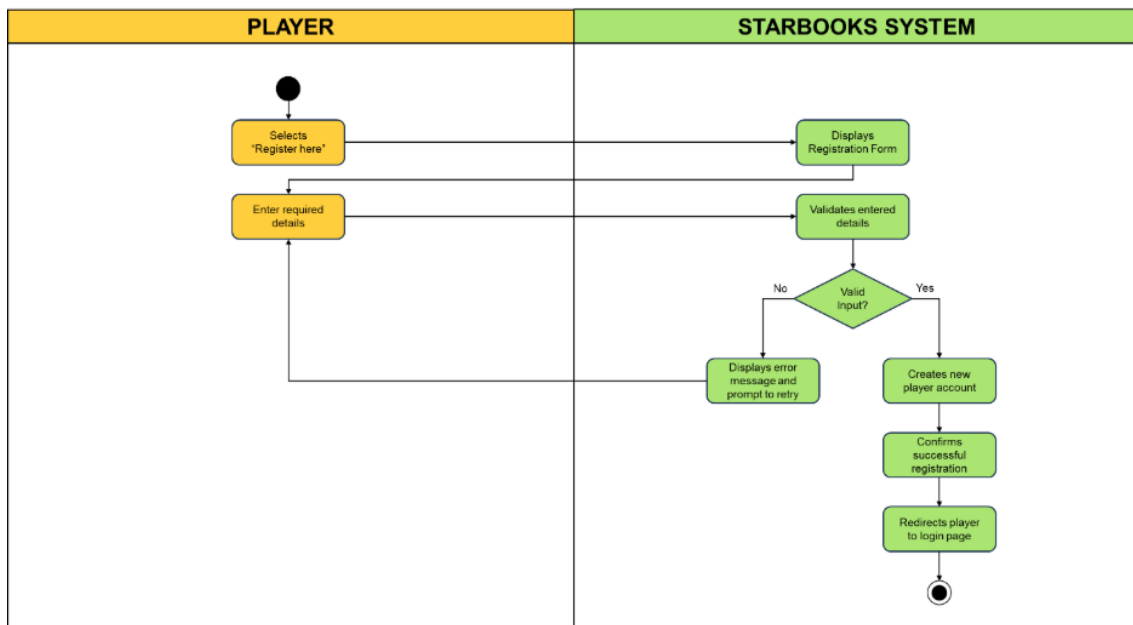
Level 2: 7.0 View Analytics Dashboard**Level 2: 8.0 View Leaderboard**

5.2 Activity Diagrams with Swim-lanes

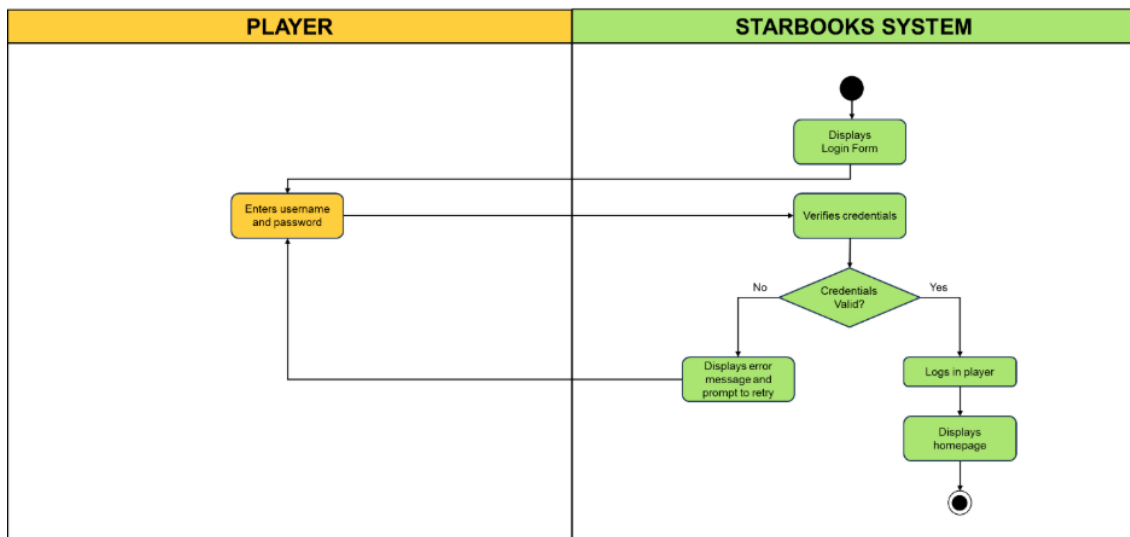
The Activity Diagrams with swim-lanes illustrate the step-by-step flow of actions within key processes of the STARBOOKS Whiz Challenge system. Each swim-lane represents a specific actor or system component, making it easy to see who is responsible for each action and how tasks move from one participant to another.

Take note that the activity diagrams included in this documentation are only those that are part of the MVP version of the application. A complete set of activity diagrams for the entire system can be found [here](#).

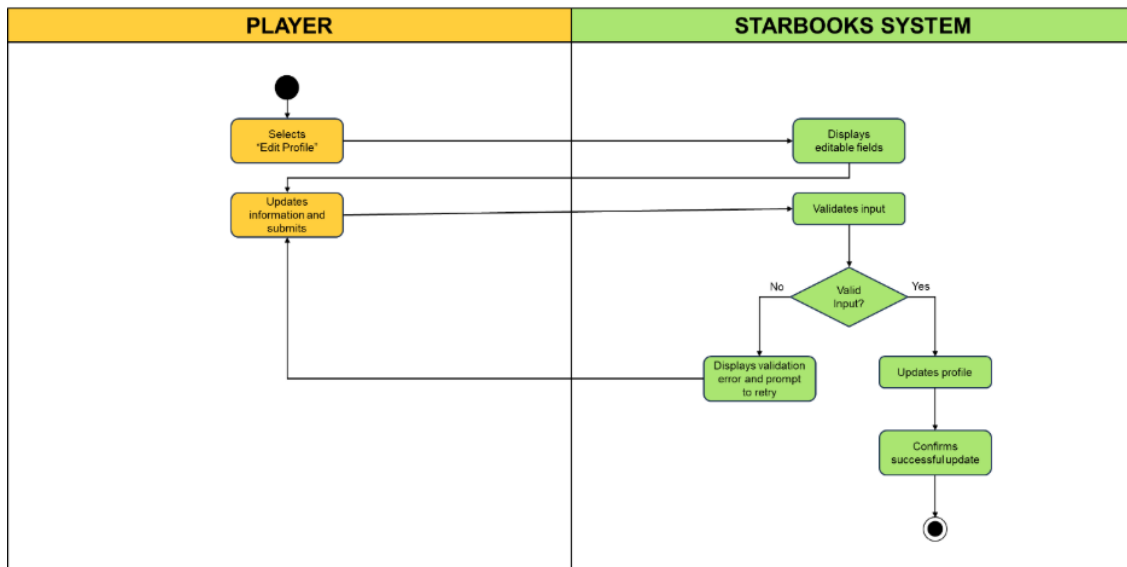
UC-MPA1: Player Register



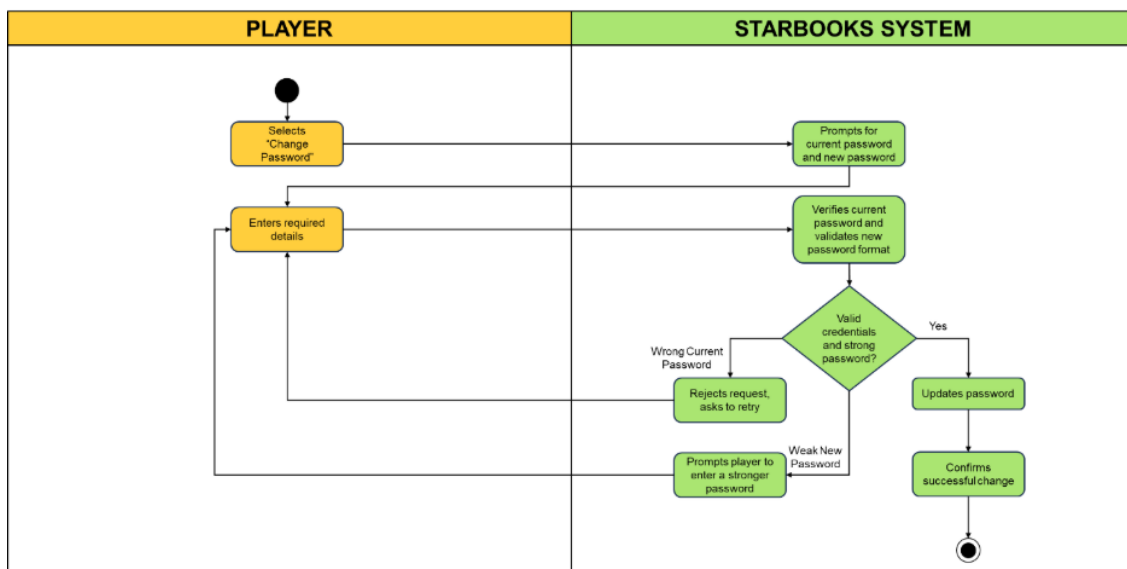
UC-MPA2: Player Login



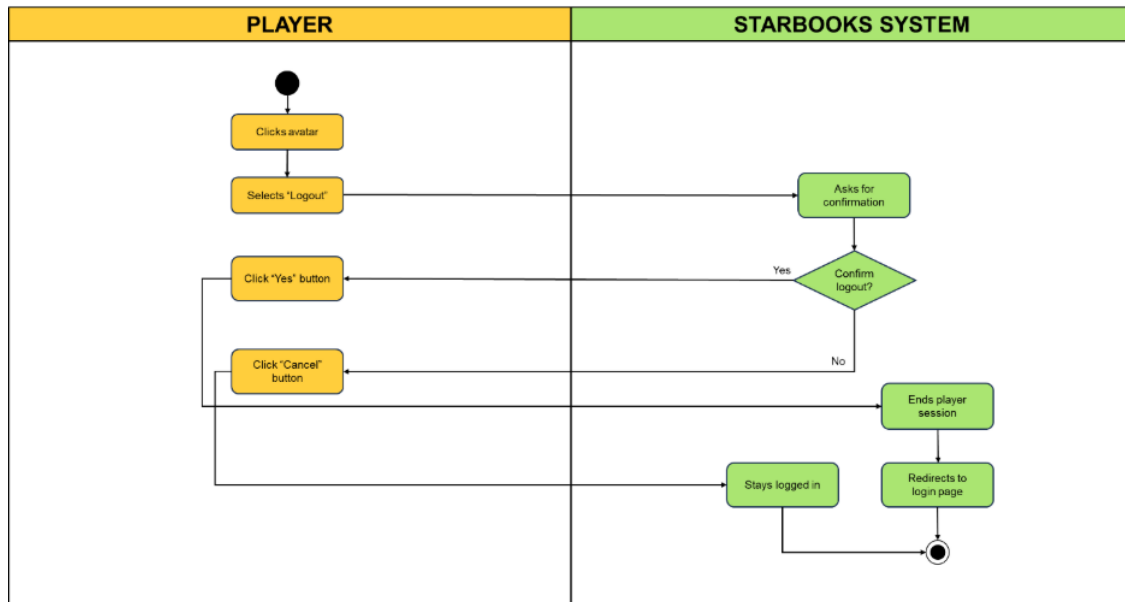
UC-MPA3: Edit Player Profile



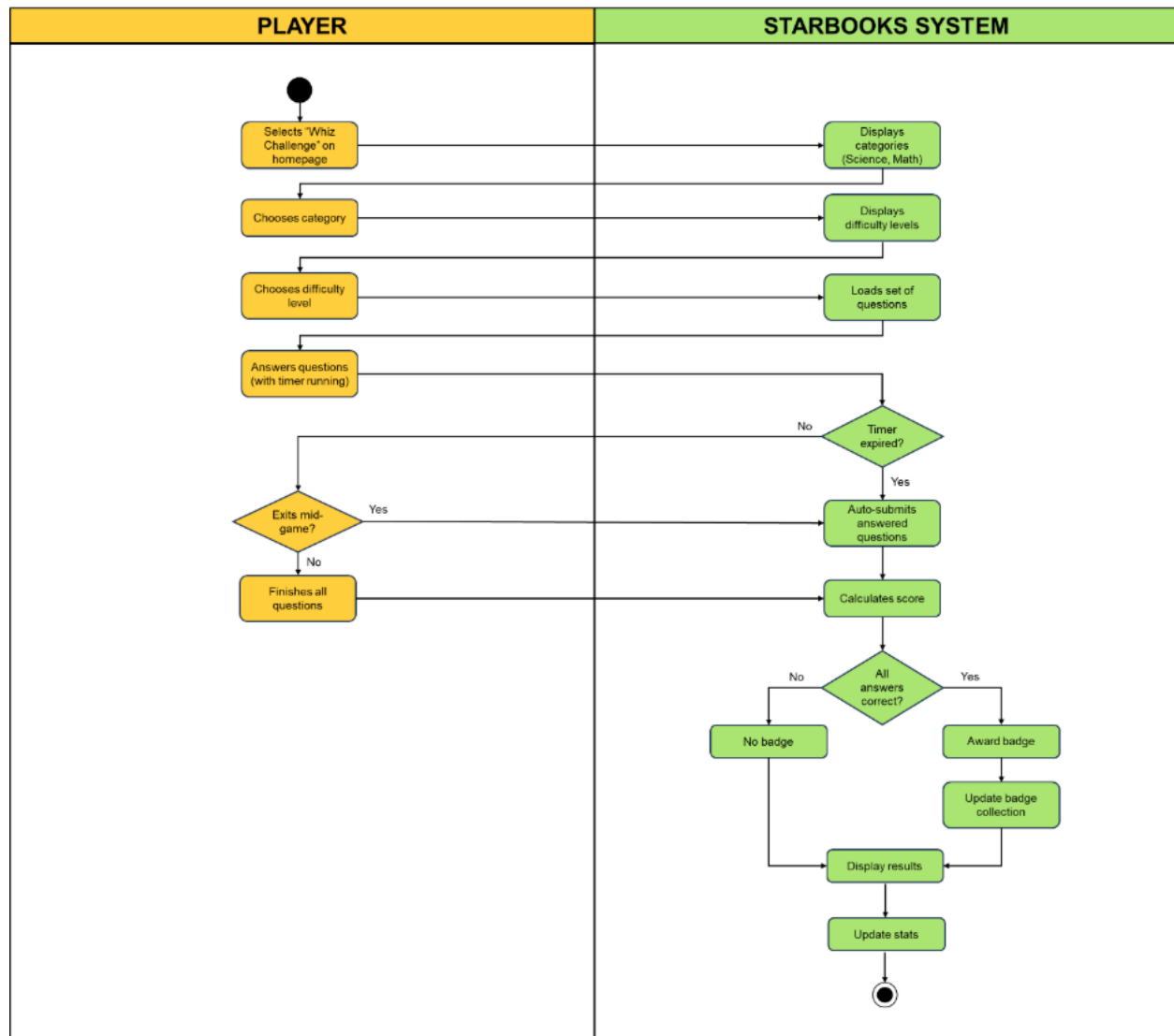
UC-MPA4: Change Player Password



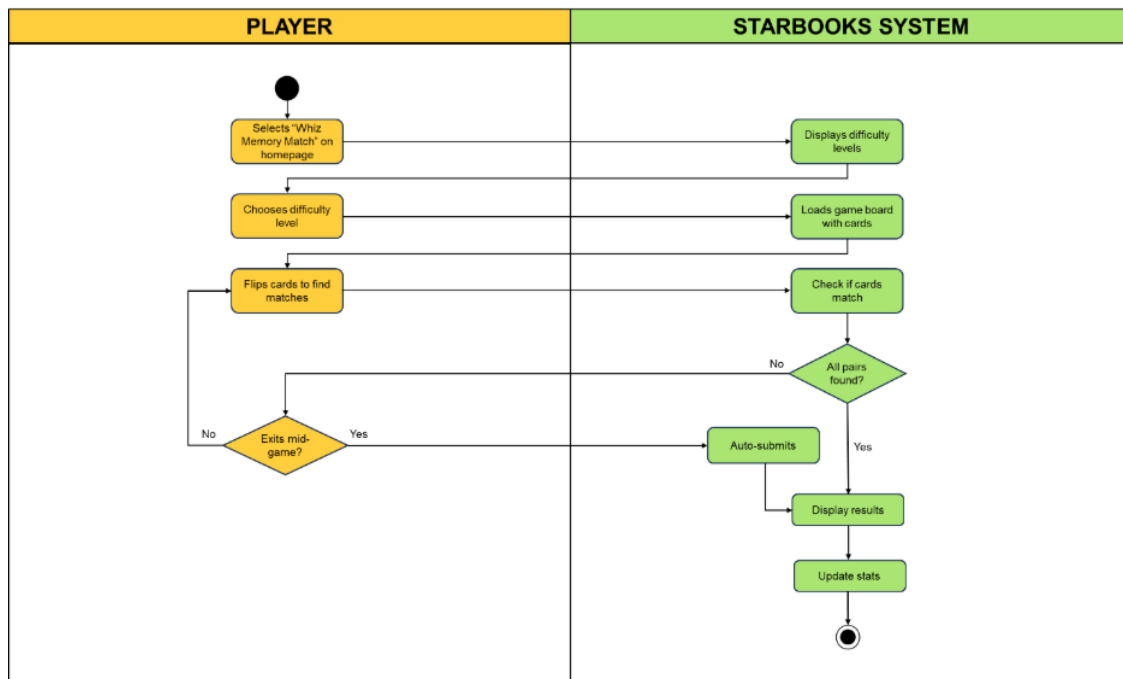
UC-MPA5: Player Logout



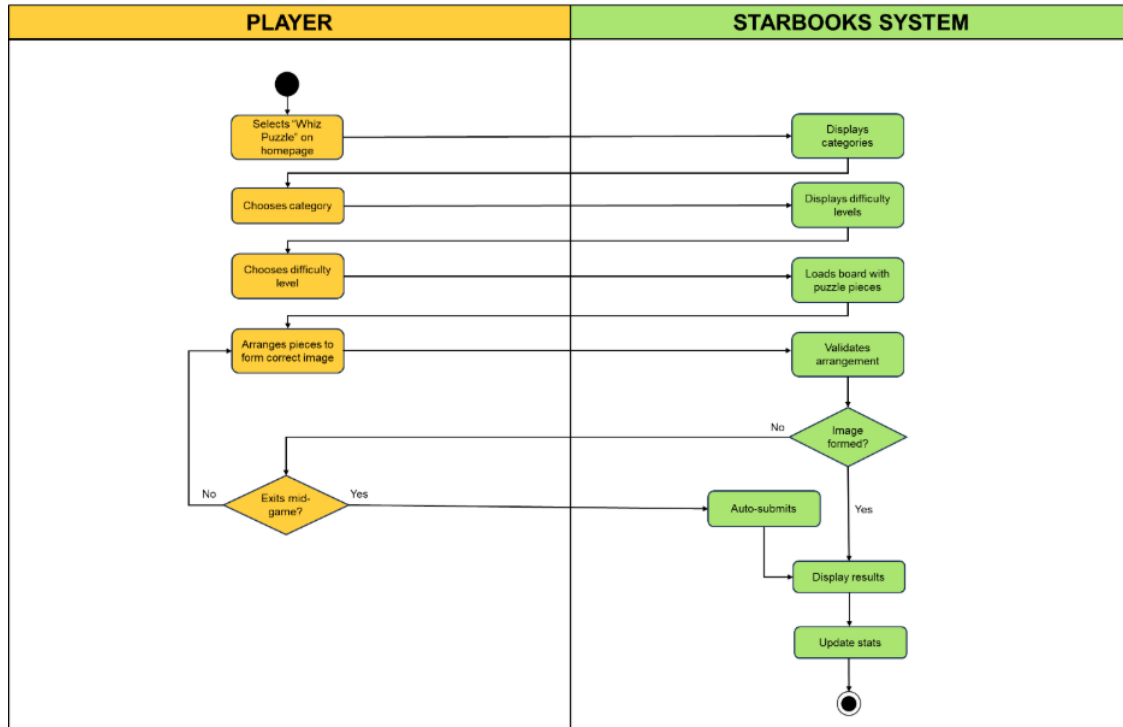
UC-PG1: Play Whiz Challenge



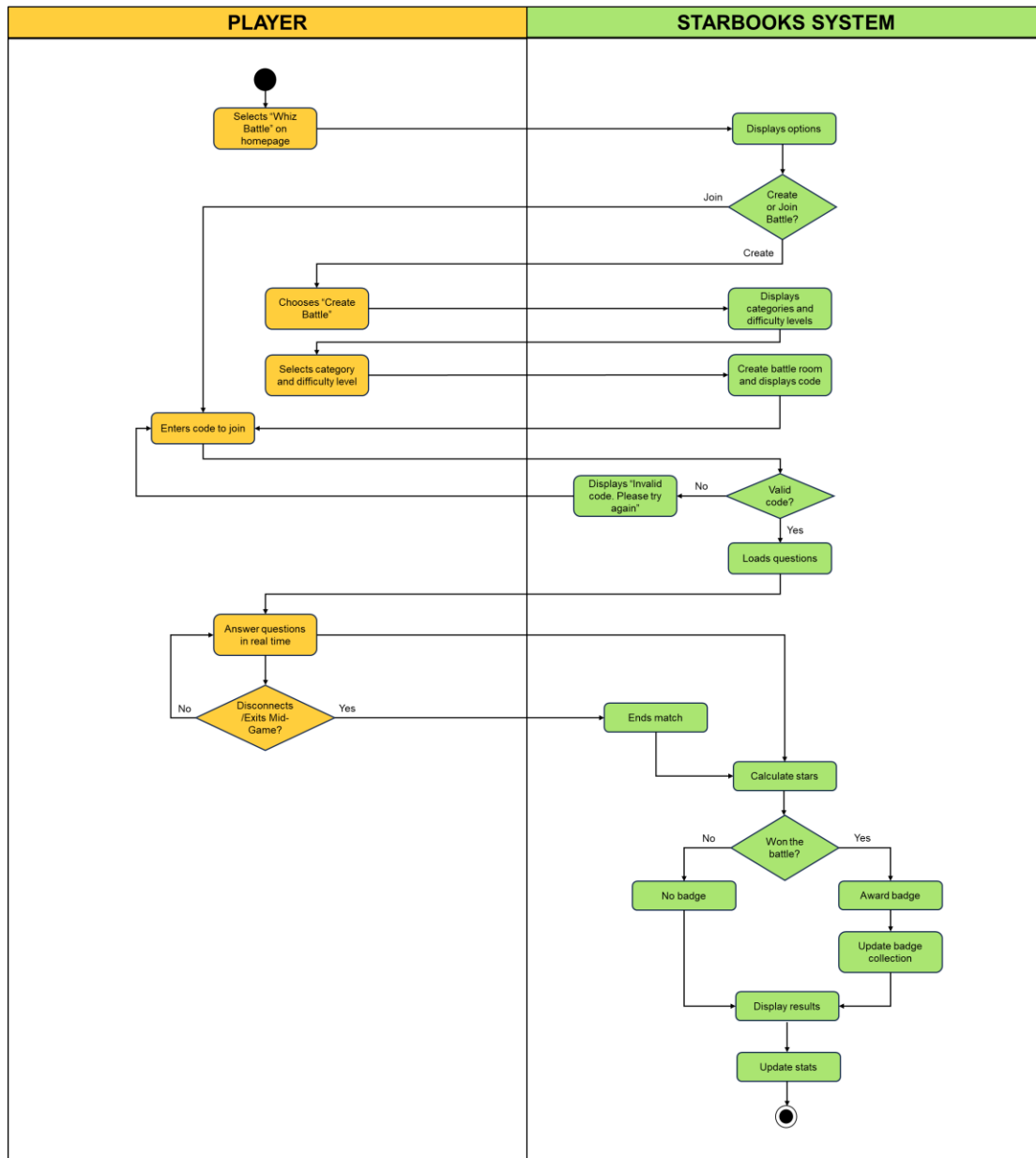
UC-PG2: Play Whiz Memory Match



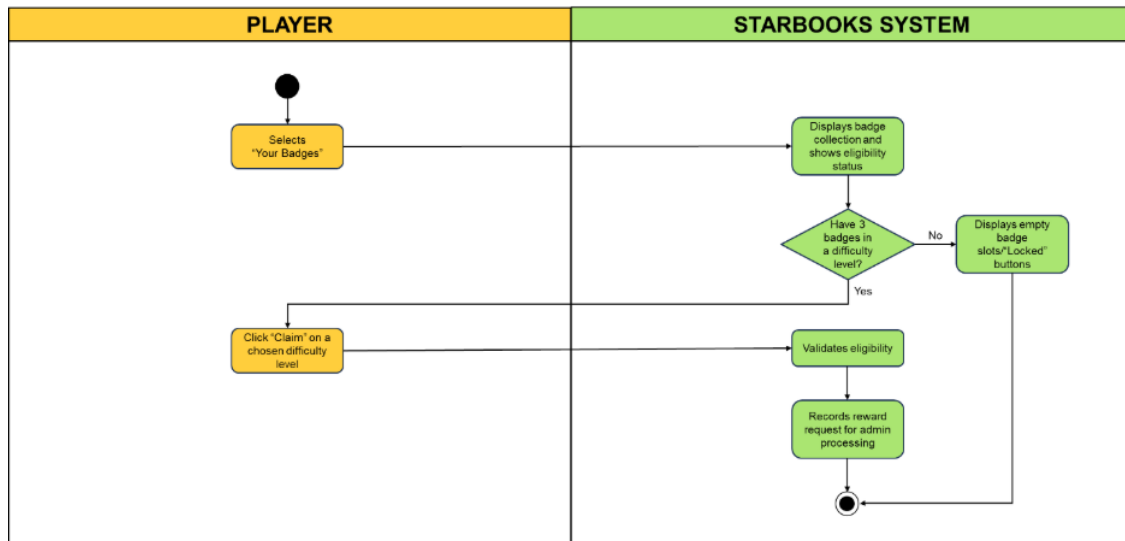
UC-PG3: Play Whiz Puzzle



UC-PG4: Play Whiz Battle



UC-RR1: Player Requests Reward

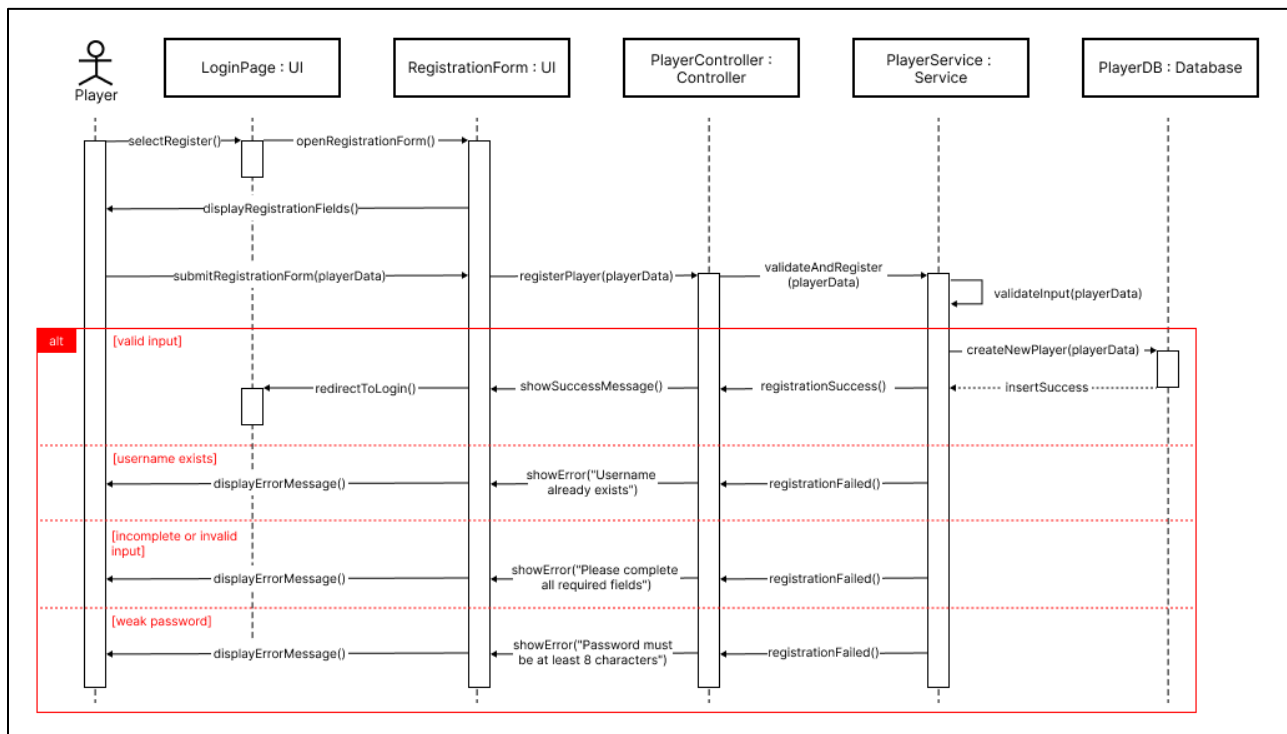


5.3 Sequence Diagrams

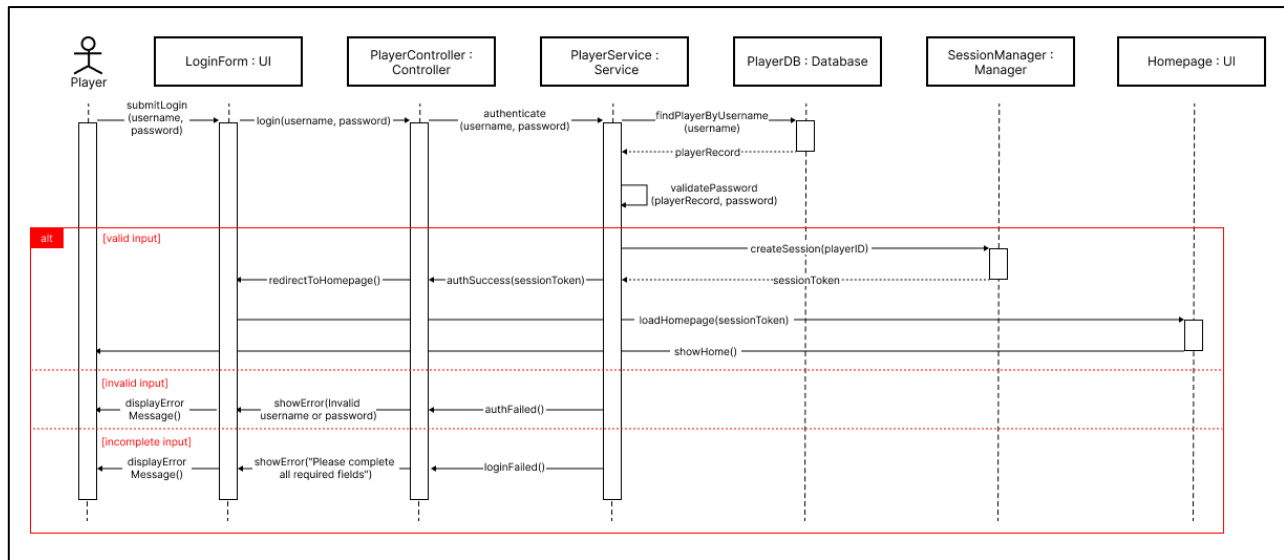
The Sequence Diagrams show how different parts of the STARBOOKS Whiz Challenge system interact with each other over time. These diagrams highlight the order of messages exchanged between actors and system components, making it easier to understand how each feature works behind the scenes.

Take note that the sequence diagrams included in this documentation are only those that are part of the MVP version of the application. A complete set of sequence diagrams for the entire system can be found [here](#).

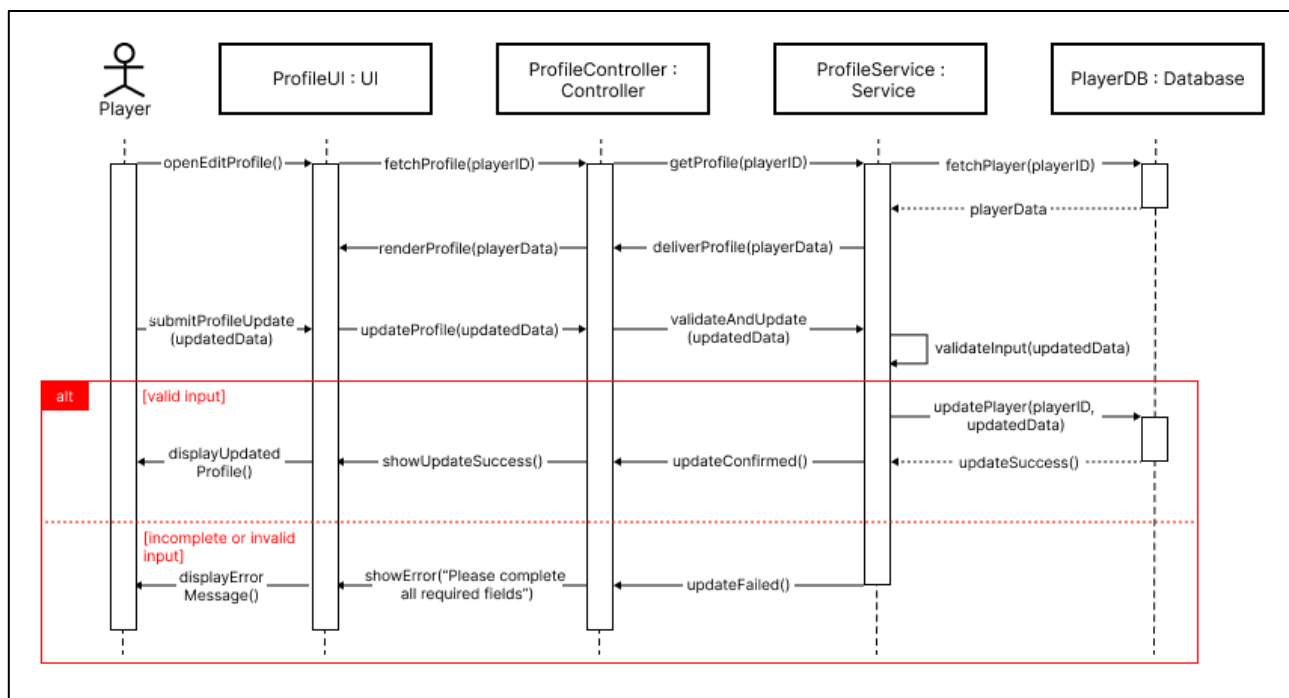
UC-MPA1: Player Register



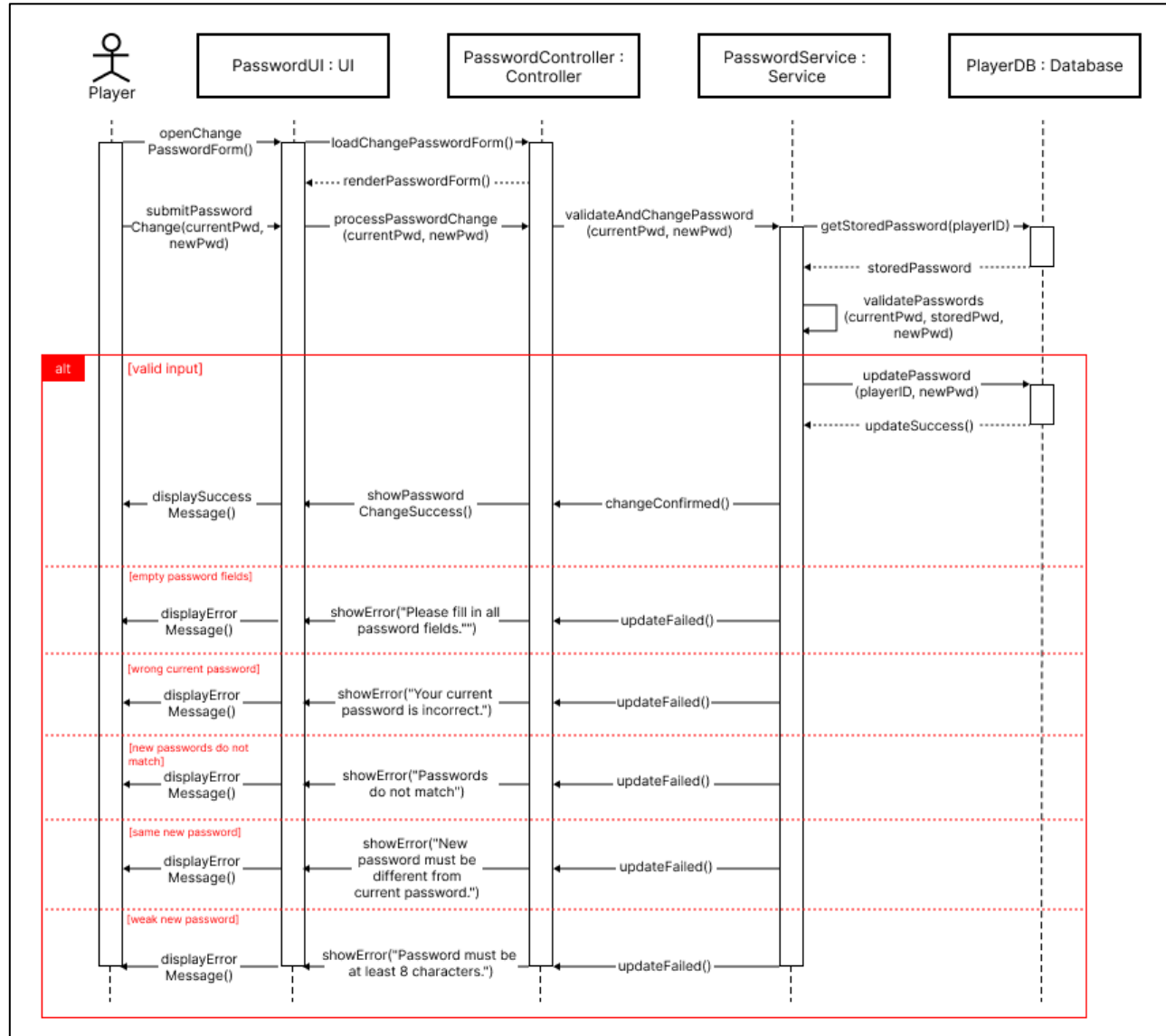
UC-MPA2: Player Login



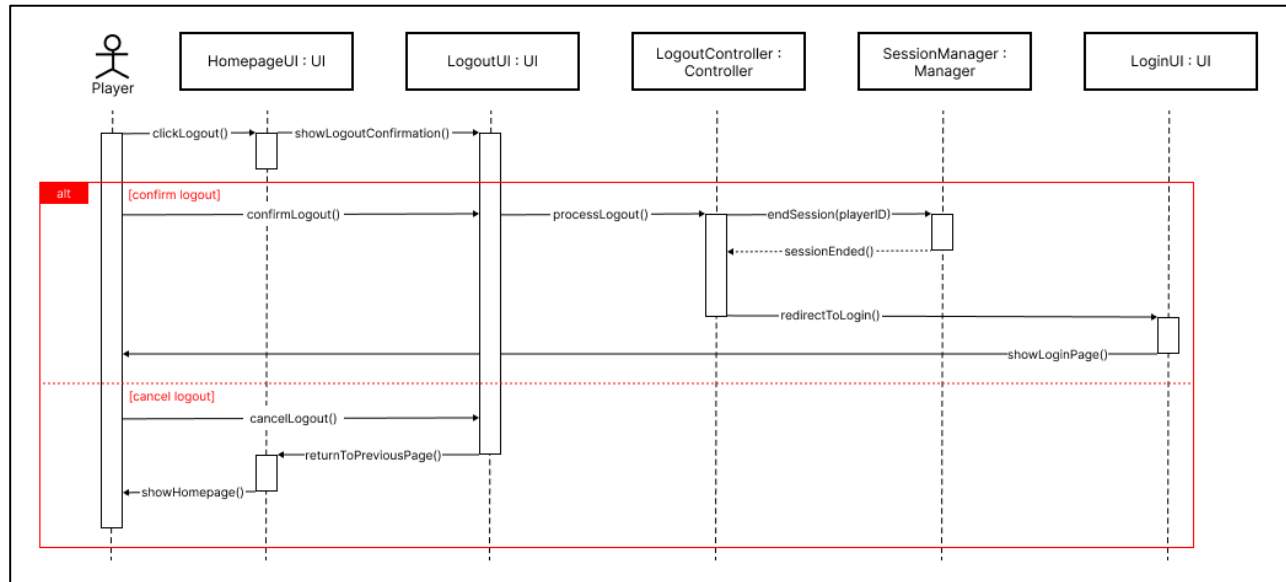
UC-MPA3: Edit Player Profile



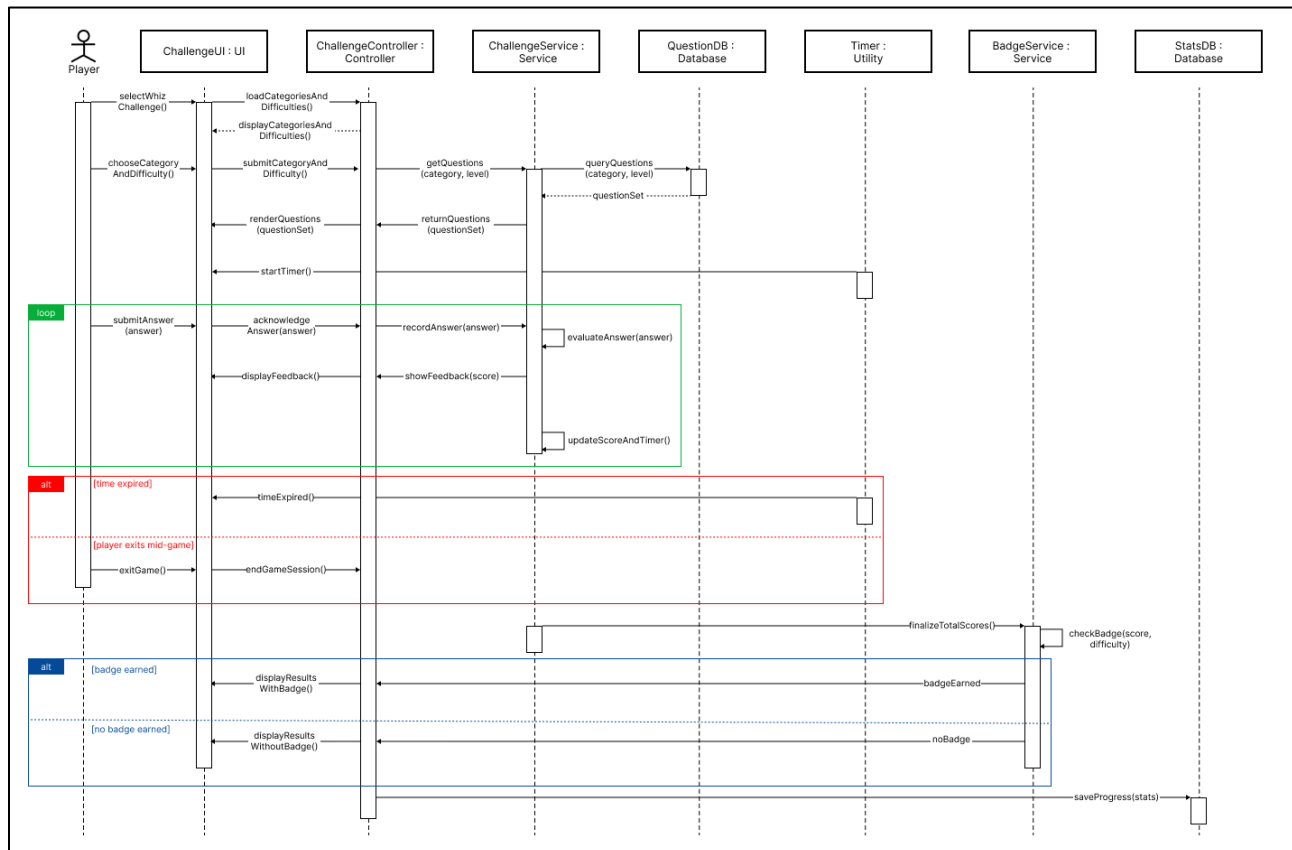
UC-MPA4: Change Player Password



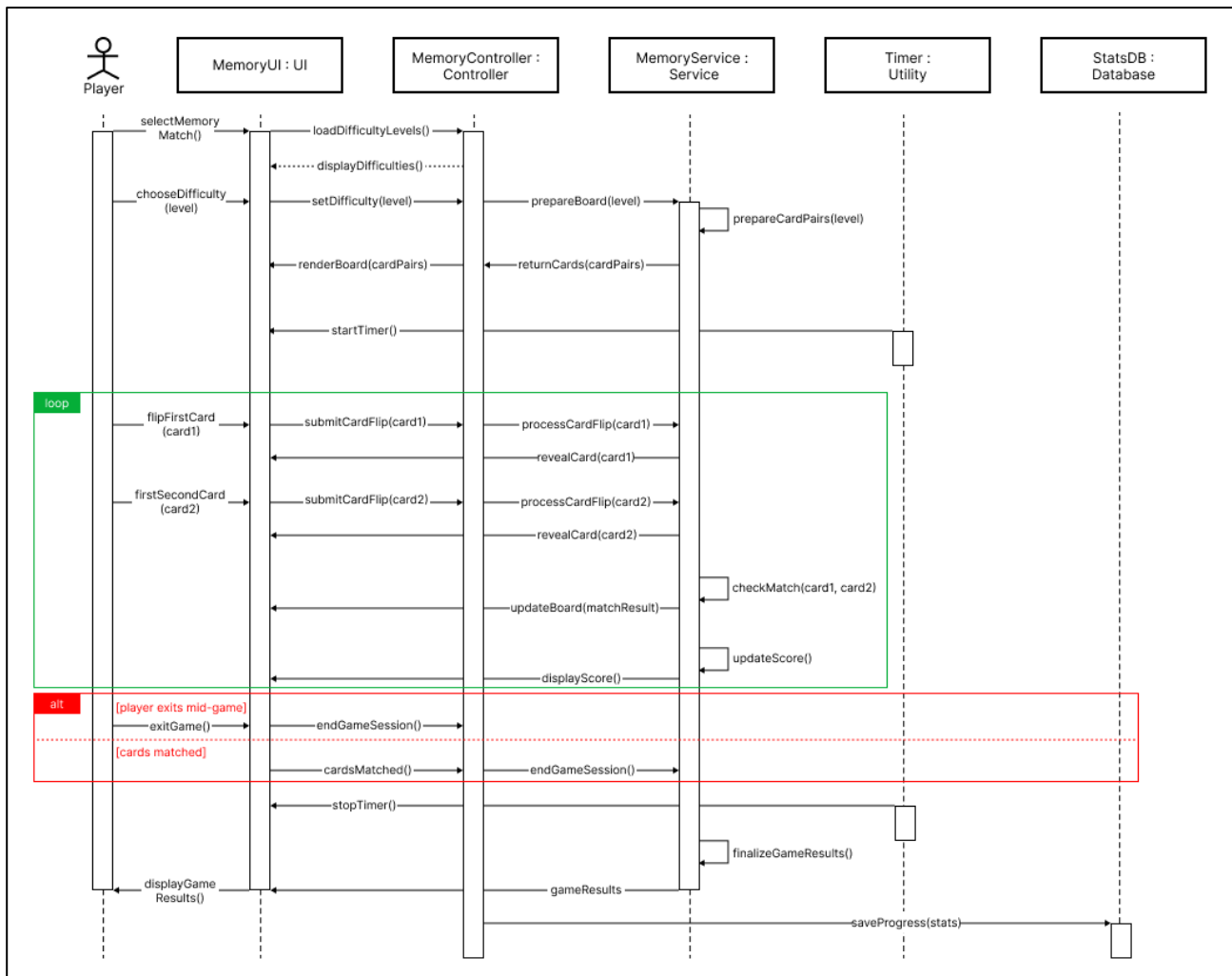
UC-MPA5: Player Logout



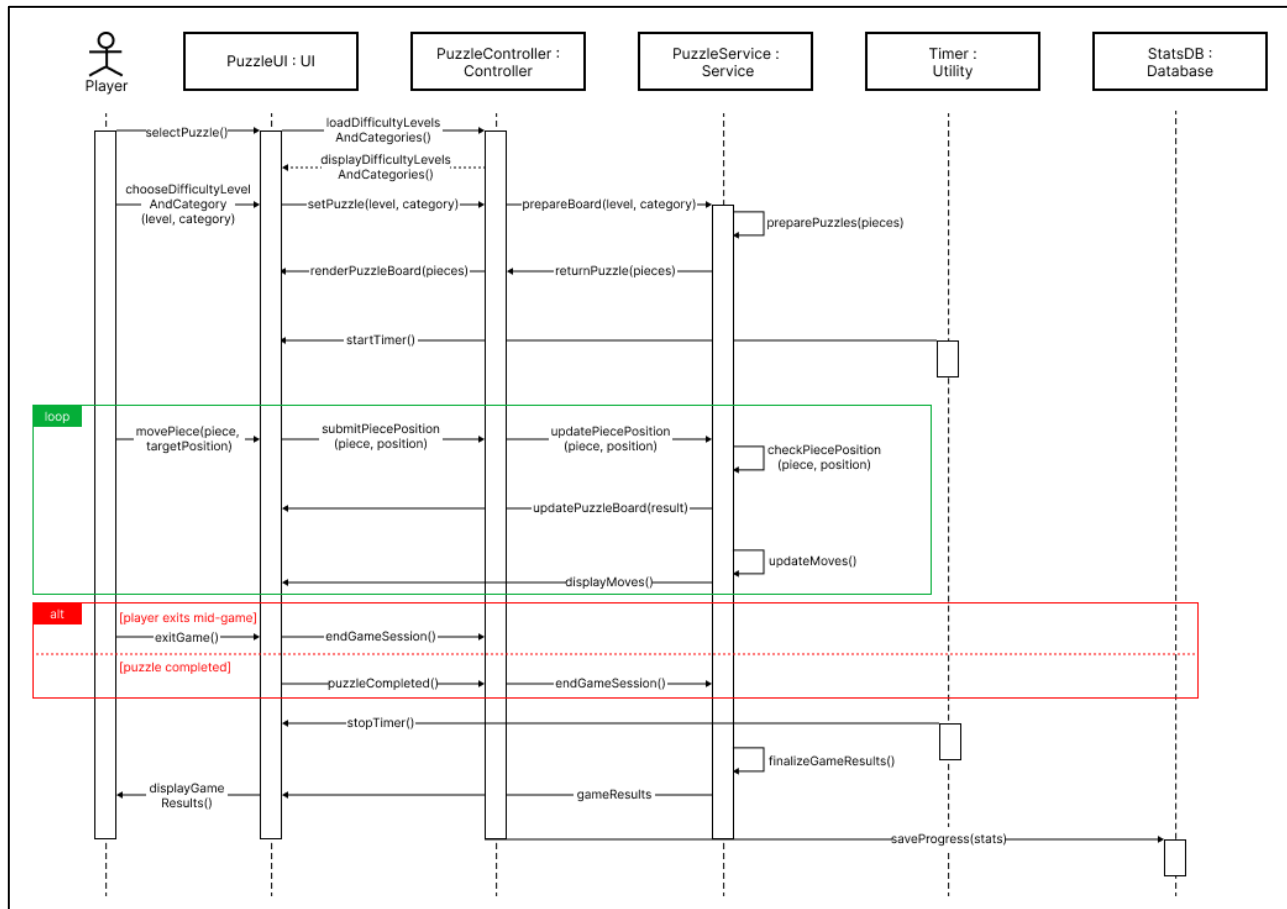
UC-PG1: Play Whiz Challenge



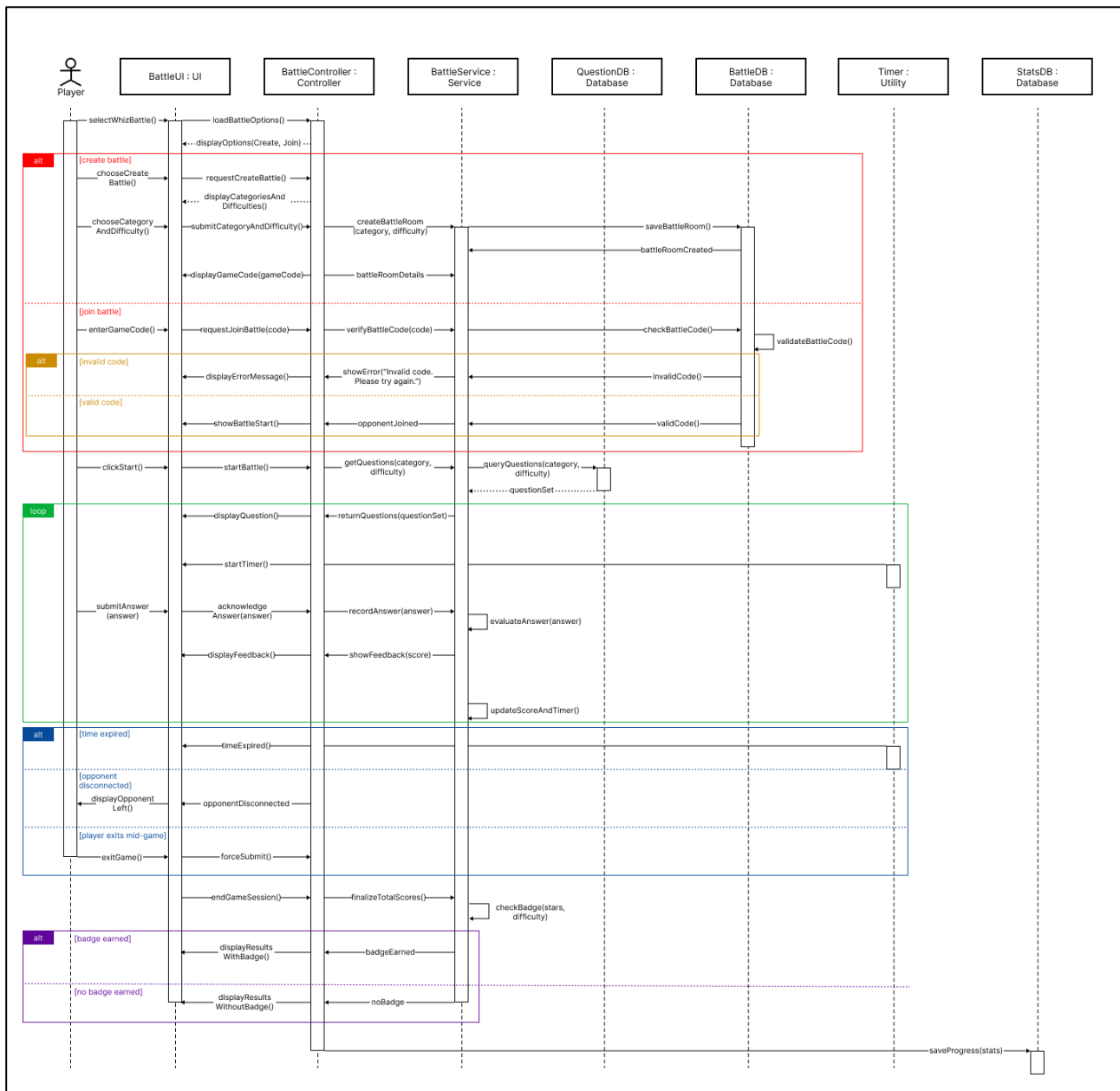
UC-PG2: Play Whiz Memory Match



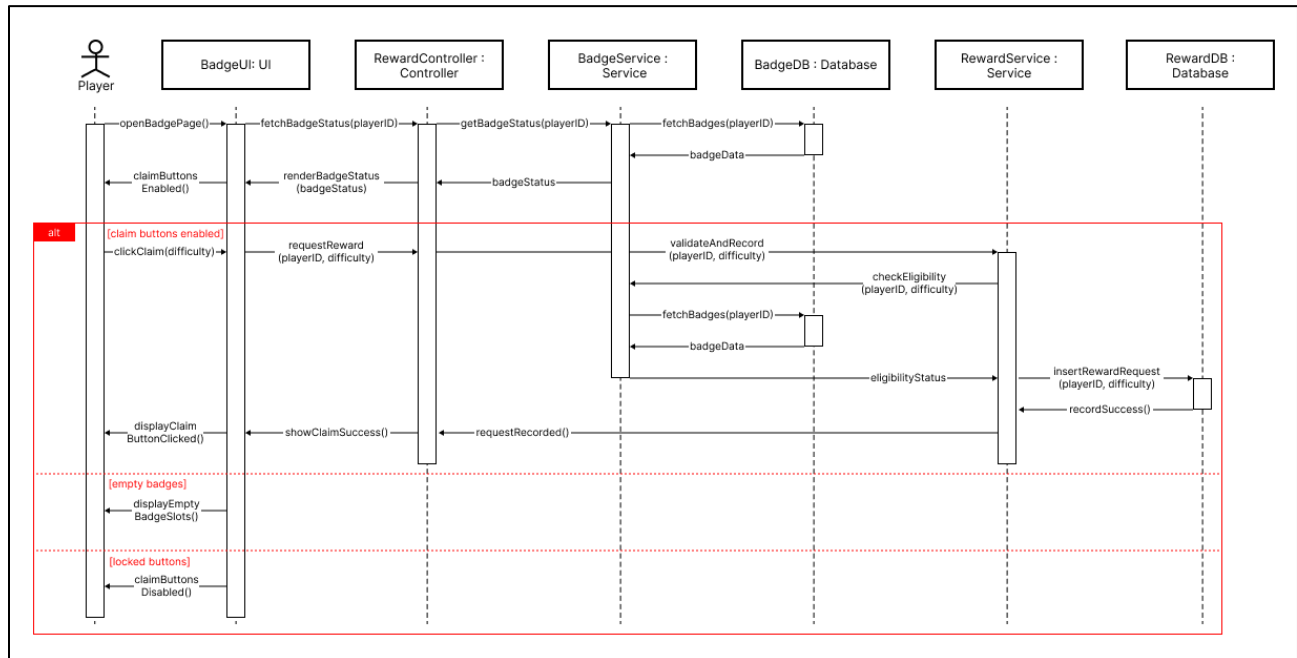
UC-PG3: Whiz Puzzle



UC-PG4: Whiz Battle



UC-RR1: Player Requests Reward

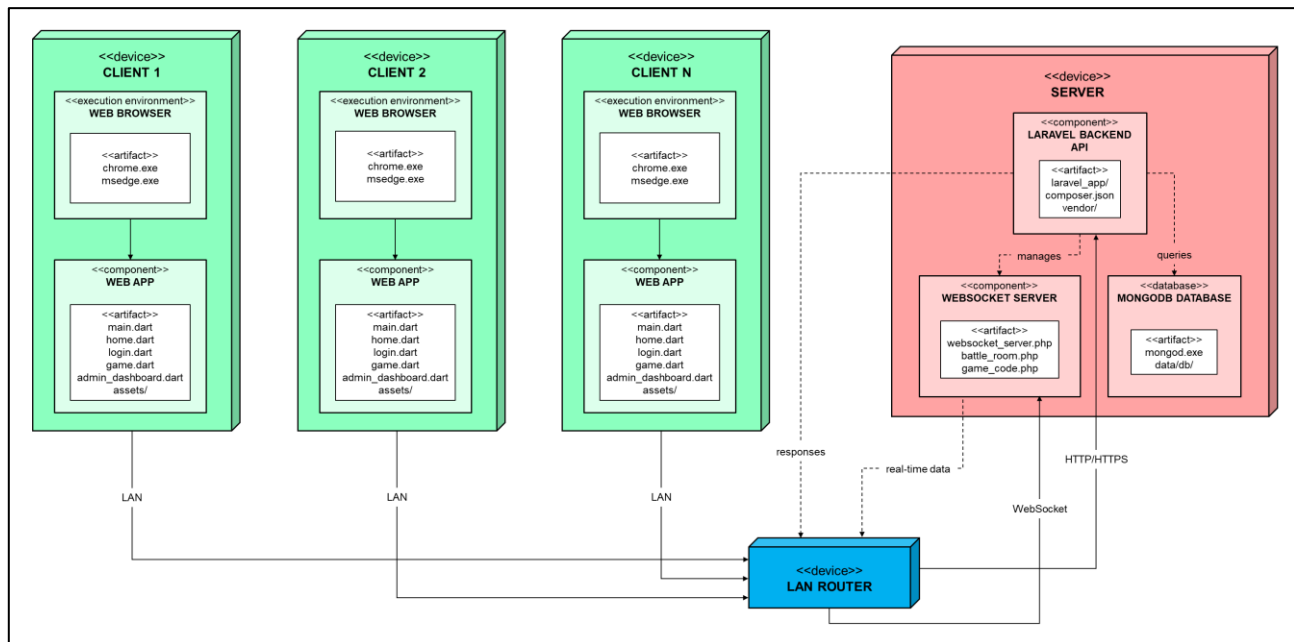


5.4 Deployment Diagram

The Starbooks Whiz system follows a client-server architecture deployed within a local area network (LAN) environment, designed to operate independently without requiring internet connectivity. The deployment consists of multiple client computers (designated as CLIENT 1, CLIENT 2, through CLIENT N) and one centralized server computer, all interconnected through a LAN Router. This architecture enables the system to be deployed in remote locations or areas with limited internet access, as all communication and data processing occur within the closed local network.

On the client side, each computer functions as a <<device>> node running a web browser that serves as the <<execution environment>> for the Starbooks Whiz Flutter/Dart web application. The browser artifacts include standard executable files such as chrome.exe or msedge.exe. The <<component>> WEB APP deployed on each client contains several <<artifacts>> including main.dart (application entry point), login.dart (authentication interface), game.dart (game modules), admin_dashboard.dart (administrative interface), and associated assets folders containing images, sounds, and fonts. Players access the four educational games through this interface, while administrators can access the admin panel from any client computer by navigating to the admin section from the login page. All client computers communicate with the server via the LAN Router using standard HTTP/HTTPS protocols for regular operations and WebSocket protocol for real-time features.

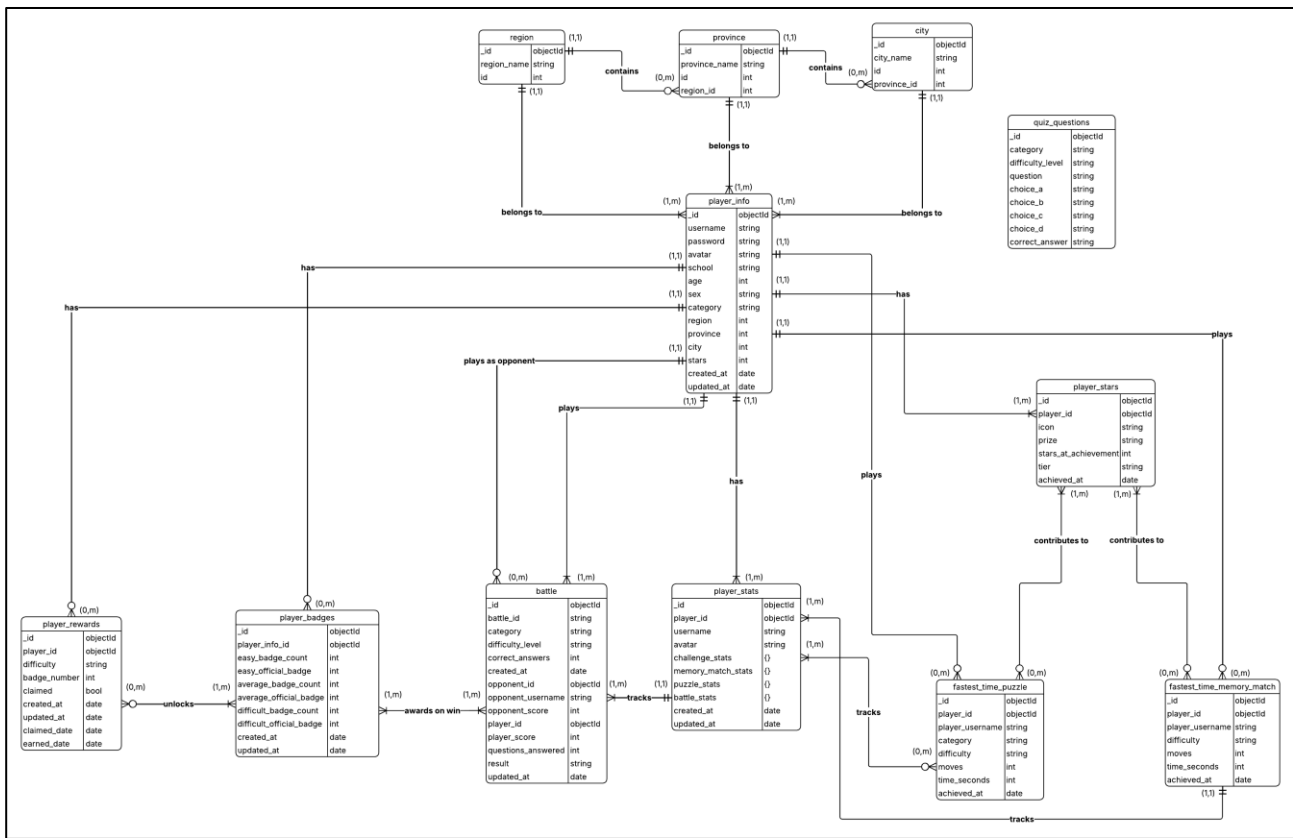
The server computer acts as the central <<device>> node hosting three primary <<components>>: the Laravel Backend API, WebSocket Server, and MongoDB Database. The Laravel Backend API, implemented as a <<component>> with artifacts including laravel_app/ directory, composer.json, and vendor/ libraries, handles all business logic, user authentication, game management, and administrative functions. The WebSocket Server <<component>>, consisting of websocket_server.php, battle_room.php, and game_code.php artifacts, manages real-time communication specifically for the 1v1 battle game mode, enabling players to create and join battle rooms using generated game codes. The MongoDB Database <<component>>, deployed with mongod.exe executable and data/db/ storage directories, persists all application data including user accounts, game scores, battle histories, and system configurations. Internal relationships within the server show that Laravel manages the WebSocket Server and queries the MongoDB Database through direct localhost connections, while external communication with client computers flows through the LAN Router.



Appendix A – Entity Relationship Diagram

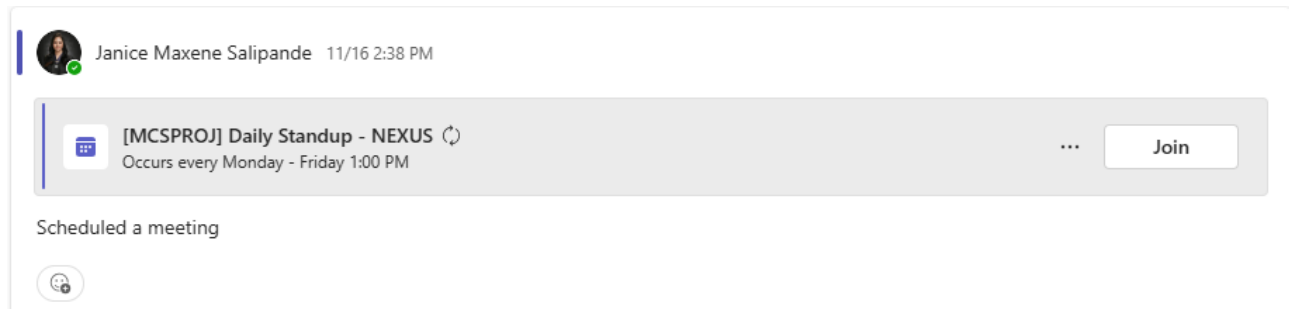
The Entity Relationship Diagram illustrates the data entities of the STARBOOKS Whiz Challenge system and the relationships between them. It shows how data is structured, stored, and connected within the system, helping clarify how different components depend on and interact with each other at the data level.

Take note that the entities included in this documentation are only those that are part of the MVP version of the application. The document containing the complete data dictionary for this ERD can be found [here](#).



Appendix B - Group Log

Daily Standup Meeting Link



Weekly Meeting with Adviser (November 18, 2025)



Date: November 18, 2025

Time: 11:30AM – 11:40AM

Attendees:

- Sir Jose Eugenio Quesada
- Kelly Dumbrique
- Arcielle Marie Gercan
- Shandrae Lois Quianzon
- Janice Maxene Salipande

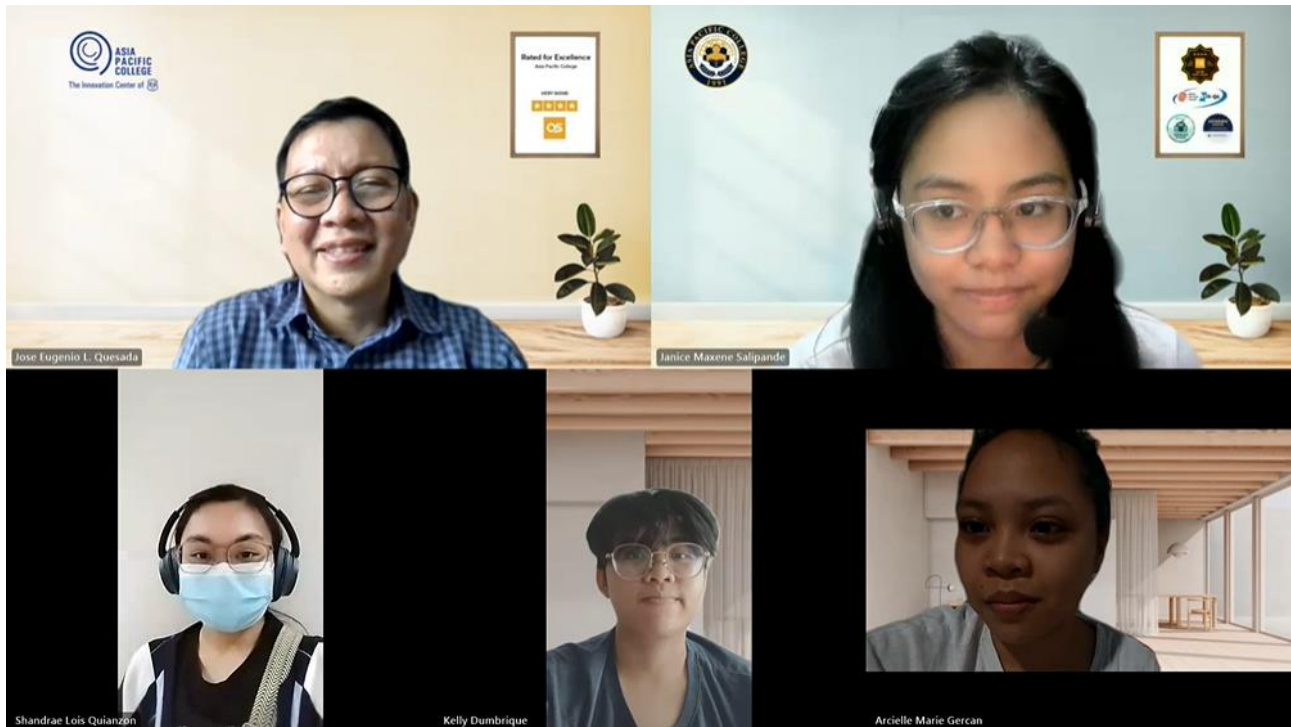
Key Discussion Points:

- **Deployment:** Discussed deployment of tools/servers.

- **Sprint Goals:** Aim to finish and present approved sprint backlog items by the end of the sprint, leading to a product increment. Highlighted the time limitation.
- **Project Timeline:** Projects span two terms (third term of the third year and first term of the fourth year). Emphasis on the need for faster progress due to prior PBL experience.
- **Questionnaire:** The provided questionnaire can be updated to be more user-friendly.

Action Items:

- Follow up DOST on the questionnaire list

Weekly Meeting with Adviser (November 25, 2025)**Date:** November 25, 2025**Time:** 11:30AM – 11:55AM**Attendees:**

- Sir Jose Eugenio Quesada
- Kelly Dumbrique
- Arcielle Marie Gercan
- Shandrae Lois Quianzon
- Janice Maxene Salipande

Key Discussion Points:**Sequence Diagrams**

- The deliverable is currently focused on sequence diagrams.
- Sequence diagrams are being created to illustrate scenarios from use case diagrams.

Sprint Progress

- Group is on Sprint 3, which ends this Friday.

Deliverables

- The midterm deliverable will be what is ready for deployment. Sequence diagrams may be included.
- Final deliverables will include what was not deployed during midterms.

Open Project

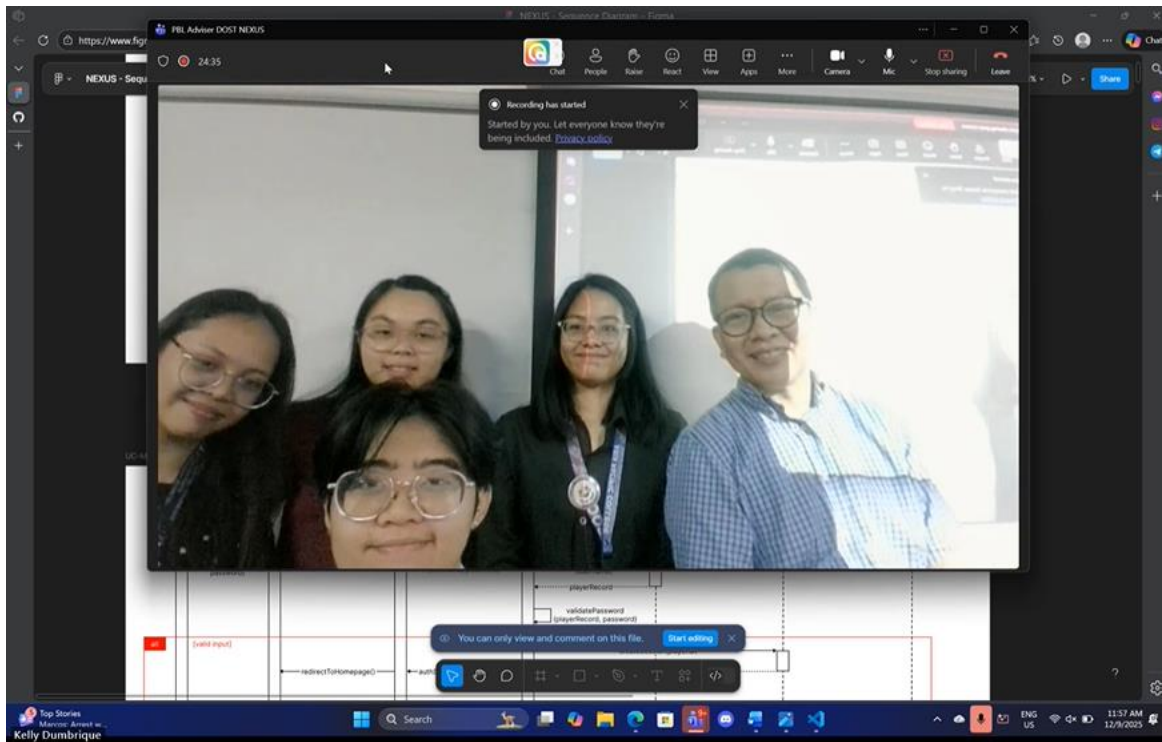
- Sir Dong is not looking at Open Project, but the team continues to use it for organization.

Future Project Considerations

- Discussed the possibility of exchanging projects between groups for a more realistic experience or continuing with the same project in the next term to implement additional features.

Action Items:

- Continue working on sequence diagrams and sprints.
- Prepare for Sprint 3 review this Friday.
- Maintain communication with the client regarding deliverables.
- Product Owner (Kelly) to manage and communicate backlog and timeline to the client.

Weekly Meeting with Adviser (December 9, 2025)

Date: December 9, 2025

Time: 11:30AM – 11:55AM

Attendees:

- Sir Jose Eugenio Quesada
- Kelly Dumbrique
- Arcielle Marie Gercan
- Shandrae Lois Quianzon
- Janice Maxene Salipande

Key Discussion Points:

- **MVP Presentation Format:** Focus should be primarily on the player side for the MVP presentation. Diagrams will be addressed later.

Action Items:

- Continue with coding and documentation

Sprint 3 Planning Meeting (November 18, 2025)

Date: November 18,2025

Time: 1:10PM – 1:30PM

Attendees:

- Kelly Dumbrique
- Arcielle Marie Gercan
- Shandrae Lois Quianzon
- Janice Maxene Salipande

Key Discussion Points:**Deployment and MVP**

- Discussion on deploying the system for user use.
- Defining the Minimum Viable Product (MVP) to prioritize development efforts.
- Diagram tasks include deployment and sequence diagrams.

Sprint 3 and 4 Planning

- Officially starting Sprint 3 from November 17 to November 28.
- Sprint board to be created.
- Final presentations for unfinished items during finals.

Development Focus

- Player account, registration, login, home page, leaderboard, and four games considered good.
- Admin dashboard needs improvement compared to Figma design.
- Assignments: Admin front end, Puzzle (front-end), backend.
- Clean up and polish existing code, focusing on footer and splash screen.

Development Workflow

- Front-end to be passed to back-end developers after completion.
- Testing of published parts (homepage, registration, etc.).
- Alternate development strategy to avoid delaying deployment of completed sections.

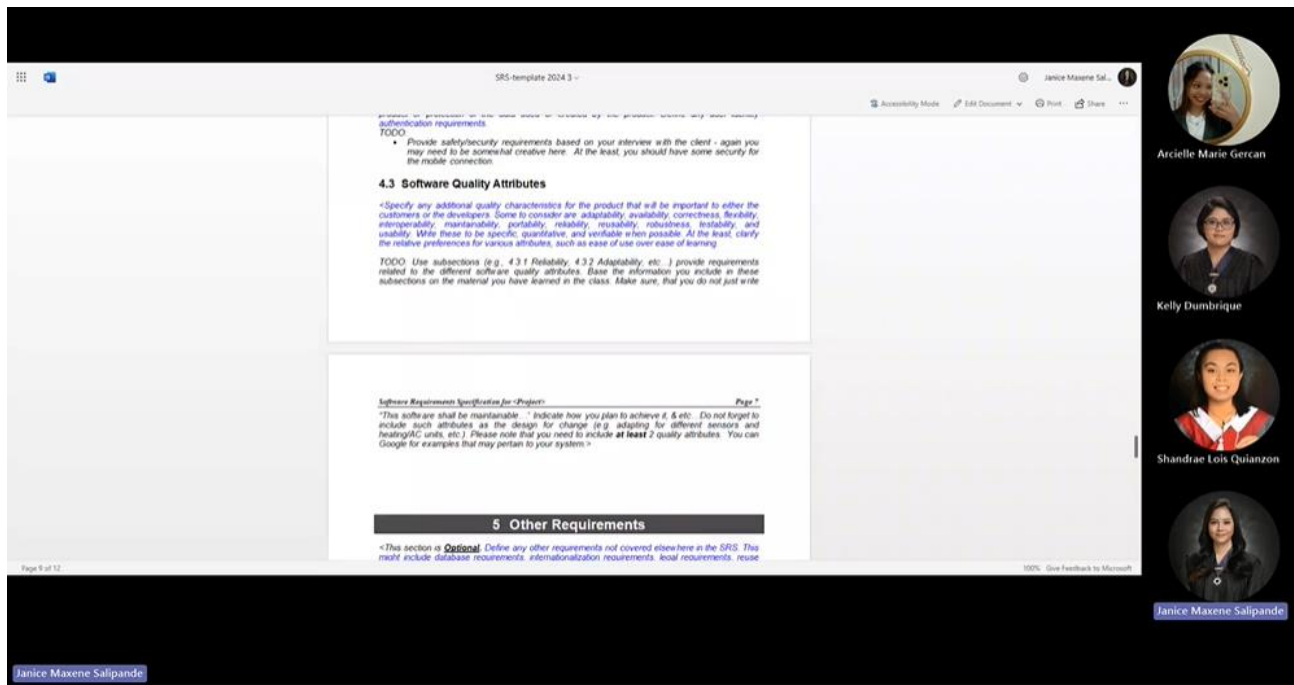
Strategy

- Arcielle and Shandre focus on polishing existing code and addressing DOST comments.

- Janice focuses on puzzle.
- Kelly focuses on admin.
- Pass front-end to back-end developers for connection.
- Test polished components.

Action Items:

- Every Tuesday meeting with Sir Joegene from 11:30 to 12:00.
- Daily stand-up meetings from Monday to Friday.
- Update project hours in OpenProject to reflect budget properly.

Group Meeting (December 5, 2025)**Date:** December 5, 2025**Time:** 1:30PM – 1:20PM**Attendees:**

- Kelly Dumbrique
- Arcielle Marie Gercan
- Shandrae Lois Quianzon
- Janice Maxene Salipande

1. Project Deliverables and Timeline

- Four weeks to finalize deliverables.

2. MVP Scope (Midterm Presentation)**2.1. Player Side**

- Accounts are functional, allowing players to play with stats.
- Missing: Whiz Battle, Leaderboard, and Badges.
- Whiz Challenge is functional (needs confirmation).
- Main Focus for Midterms: Player side functionalities.
- Leaderboard should work with multiple players.
- Badges system (without rewards) should be functional for the leaderboard.
- Claim functionality is not yet applicable/ready.
- MVP Status (Player Side): Estimated at 70% completion. Primary focus

2.2. Admin Side

- Kelly has completed the frontend.
- If extra time permits, show the frontend of the admin side.

3. Tasks and Responsibilities

- **Kelly and Janice:** Focus on SRS (System Requirements Specification).
- **Arcielle and Shandrae:** Continuing current tasks.

4. SRS (System Requirements Specification)

- Need to locate previous SRS from upper batch for reference.
- Revise MNTSDEV and MSYADD final docs; include diagrams.
- Include system requirements, use cases, and visual scope documents.

5. OpenProject

- Adjust dates in OpenProject.
- Log working hours for budget visibility.

6. Scrum Report and Sprint Review

- Last day before holiday: December 19th.
- Schedule sprint review with DOST (December 16-17).
- Schedule sprint review with Sir Joegene (flexible).
- Sprint retrospective required for formality.
- Arcielle to aim for Badges and Leaderboard around December 15-19.
- Kelly to message DOST for availability between December 15-19 for sprint review.

7. Group Log and GitHub Repository

- Include minutes from major meetings in the group log.
- Add the MCSPROJ folders to the GitHub repository.
- Individual commits.
- Update SharePoint folder.

8. Testing

- Mark test cases as tested for the midterm presentation.
- Testing will be required in the first week of January.
- Meetings on January 2 - 5 to finalize presentation materials, commits, etc.
- Finalize documentation with links to files.

9. MVP Summary

- MVP Focus: Player side (70% complete).
- Exclude admin side from the MVP.
- Focus on Badges, Leaderboard, and potentially Whiz Battle.
- Continue testing.

Action Items:

- Kelly: Contact DOST for sprint review scheduling (Dec 16-17).
- Arcielle: Test leaderboard with multiple players.
- Kelly: Locate previous SRS documents.
- Janice: Update OpenProject dates and log hours.
- Janice: Add folders and update GitHub repository.
- Schedule meetings for January 2-5 for finalization.

Sprint 4 Review Meeting with Adviser (December 16, 2025)



Date: December 16,2025

Time: 11:30AM – 12:00PM

Attendees:

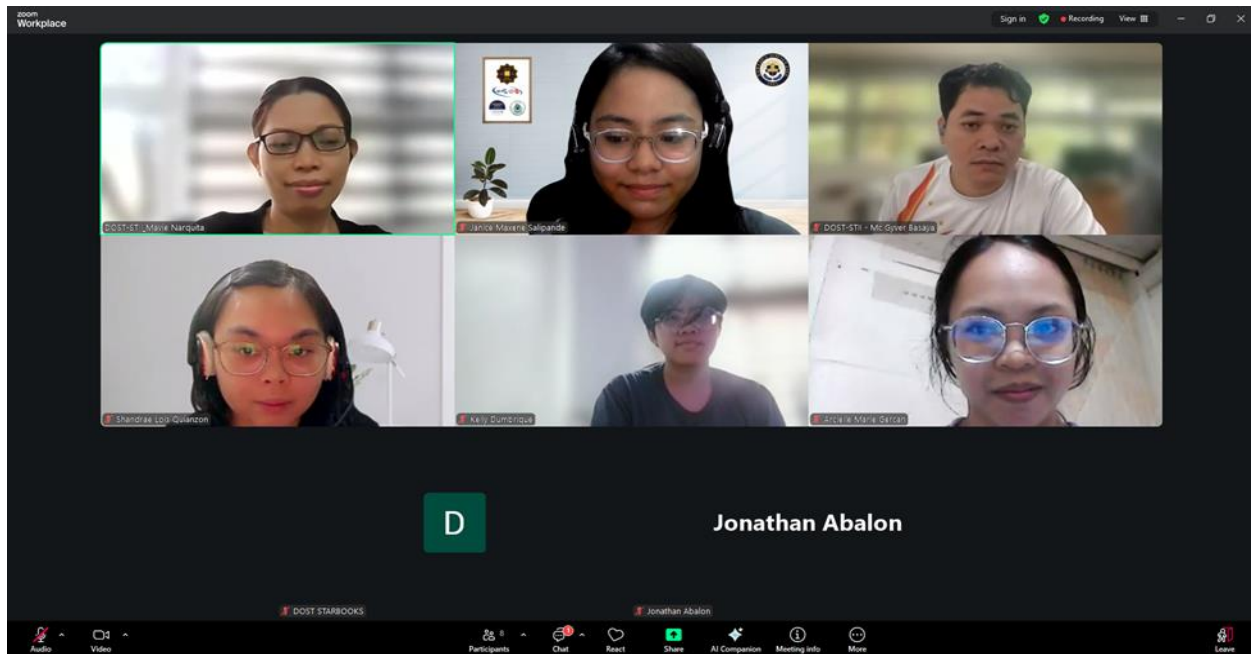
- Kelly Dumbrique
- Arcielle Marie Gercan
- Shandrae Lois Quianzon
- Janice Maxene Salipande
- Sir Jose Eugenio Quesada

1. MVP Summary

- MVP Focus: Player side (70% complete).
- Continue testing.

Action Items:

- Follow up Sprint Review with DOST
- Continue testing
- Prepare for defense

Sprint 4 Review Meeting with DOST (December 17, 2025)

Date: December 17, 2025

Time: 10:30PM – 11:30PM

Attendees:

- Kelly Dumbrigue
- Arcielle Marie Gercan
- Shandrae Lois Quianzon
- Janice Maxene Salipande
- Marie Narquita
- Mc Gyver Basaya
- Jonathan Abalon

1. Timeline and Objectives:

- The team is currently in Development Phase 2.
- The meeting's objective is to demo the player side of the application.
- Midterm presentation is scheduled for January 6 (next year), focusing solely on the player side.

2. Player Side Features:

- Login and Registration
- Homepage
- Profile
- Four Games:
 - Whiz Challenge
 - Whiz Memory Match
 - Whiz Puzzle
 - 1v1 Battle (primarily front-end, game code synchronization in progress)

3. Development Plan (November 2025 - March 2026):

- Complete 1v1 battle.
- Polish the leaderboard.
- Begin development of the admin side (dashboard, analytics).
- Testing and Finalization
- SRS (Software Requirements Specification) document completion by June 2026.

4. Demo and Discussion:

- **Login/Registration:**
 - Username, school, age, avatar, gender, and region selection.
 - Province and city selection are functional but experiencing loading delays in the local environment.
- **Homepage:**
 - Displays the four games.
- **Edit Profile:**
 - Allows users to change username, school, age, region, province, and city.
 - Change password functionality is implemented.
- **Whiz Memory Match:**
 - Minor UI fixes are needed.
 - Easy (5 pairs), Average (6 pairs), and Difficult (7 pairs) levels.
 - Cards are randomized upon game restart.
 - Fastest time recorded for both the player and all players.
- **Whiz Puzzle:**
 - Difficulty levels: Easy, Average, and Difficult.
 - Categories: Solar System Science, Human Body, Animal, Geometry, and Starbooks.
 - Images are royalty-free (Starbooks images sourced from their Facebook page).
- **Whiz Challenge:**
 - Functional with category and difficulty settings.
 - Questions are randomized based on the provided question database.
 - A pause is implemented after each question.
 - Earned badges are displayed upon completion (connected to the database).
 - Badges are awarded based on completing sets of 1-10 questions perfectly (up to three badges).
 - Claim button functionality is implemented and records the claim time.
- **1v1 Battle:**
 - Still a Work in Progress (WIP), UI is for testing purposes only.
 - Errors exist where questions are not the same for both players.
 - Opponent names are not always displayed correctly.
 - WebSocket integration is implemented for real-time battle rooms.
 - Unique six-code battle codes are generated.
 - Victory and defeat screens are implemented.
 - Scoring system: 15 points total in 15 seconds; points are awarded based on remaining time.

5. Feedback and Suggestions:

- **Background Effects:** Slow down background effects to reduce distraction/eye strain.
- **Server Setup:** Explore setting up a virtual machine for testing deployment.
- **Whiz Challenge Question Area:** Reduce the size of the choice areas.
- **Questions with images:** Implement them.
- **Game Star Rewards:** Consider implementing a star system for Whiz Memory Match and Whiz Puzzle, where stars can be earned based on completion time or difficulty.

6. Leaderboard and Admin Side:

- Leaderboard is a work in progress due to UI and API connection errors
- The admin dashboard has a UI, but it is hardcoded with persisting errors.

7. Development Process:

- The current build is considered "Version 1."
- Focus is on deploying essential features first.
- Future iterations (Version 2, Version 3) may include additional features like categories for Whiz Memory Match, medals for Whiz Battle, or expanding the 1v1 Battle player capacity.
- The Admin interface will be deployed separate to the user interface.

8. Questions and Technical Specifications:

- **PC specifications for deployment:**
 - Base configurations on internal testing setups.
 - Minimum Requirements is at least Windows 10 or later version.
- **Number of PCs for events:**
 - Two interactive kiosks will be used.
- **Simultaneous users:**
 - Recommend limiting to a maximum of 5 simultaneous users per server, when using local Wi-Fi connection, to optimize performance.

9. Interactive Kiosk Specs

- **Operating System:** Windows
- Admin dashboard will be accessible only to DOST staff.
- **Web Browser:** Edge (recommended), developed in Edge but compatible with Google Chrome.

10. Deployment Diagram:

- Deployment diagram was reviewed. It was based on a model and framework using the shown components.
- The deployment diagram to be shared to the IT Department.

11. Action Items:

- Share screenshots and the deployment diagram.
- Share the SRS document for feedback.
- Share any models and frameworks used to construct the deployment diagram.

Sprint 4 Retrospective Meeting (December 19, 2025)

The screenshot shows a Miro board titled "Sprint 4 Retrospective - NEXUS" by Janice Salipande. The board is organized into three columns: "Glad" (Positive experiences), "Sad" (Negative experiences), and "Mad" (Frustrations and challenges). Each column contains sticky notes with feedback from anonymous users, including titles, descriptions, and vote counts.

Glad (Positive experiences):

- Finish tasks on time** (4 Votes): "We're on track with our development based on our project roadmap" (4 votes).
- Code Works!** (0 Votes): "Code Works!" (0 votes).
- Good communication within the team, adviser, c** (1 Vote): "Good communication with members, client, and adviser" (1 vote).

Sad (Negative experiences):

- Project scope getting changed** (3 Votes): "Client Demands" (2 votes).
- Code errors** (1 Vote): "Being unable to resolve code errors within my set time" (1 vote).
- Continuous revisions on documentation/diagn** (4 Votes): "A lot of revisions for diagrams" (4 votes).
- Wasn't focused with tasks due to other requirements** (0 Votes): "Wasn't focused with tasks due to other requirements" (0 votes).

Mad (Frustrations and challenges):

- Claude limit** (0 Votes): "Claude limit" (0 votes).
- Code errors** (2 Votes): "Code Errors" (2 votes).
- Project scope getting changed** (1 Vote): "Our scope is getting bigger and bigger, as every meeting the client adds more features" (1 vote).
- Sometimes our client replies late when we have questions or asking for something** (0 Votes): "Sometimes our client replies late when we have questions or asking for something" (0 votes).

A "Feedback" sidebar is visible on the left, and a "Ready! 4/4" status indicator is at the top.

Date: December 19, 2025

Time: 9:37AM – 10:02AM

Attendees:

- Kelly Dumbrique
- Arcielle Marie Gercan
- Shandrae Lois Quianzon
- Janice Maxene Salipande

1. Template: Used Glad, Sad, Mad in Kollabe

2. Glad (What Went Well)

- Team consistently met deliverable deadlines for both development and documentation
- Strong communication maintained across team members, adviser, and client
- Successfully debugged and resolved multiple technical issues to achieve working functionality

3. Sad (What Could Be Better)

- Scope creep during client meetings—new features continuously added without adjusting timeline or priorities
- Multiple rounds of diagram revisions delayed development phase, despite team believing documentation was complete
- High volume of code errors impacted sprint velocity

4. Mad (What Frustrated Us)

- Significant time lost troubleshooting unexpected code errors that blocked progress

5. Action Items

- Maintain OpenProject tracking and Scrum methodology for development and documentation visibility
- Keep daily standups and scheduled adviser/client meetings
- Present MVP concept to client with clear feature prioritization
- Implement a code review process before merging changes into the main branch
- Allocate extra time specifically for fixing bugs, errors, and unexpected issues

Appendix C – Test Cases

The Test Cases section outlines how the STARBOOKS Whiz Challenge system will be evaluated to ensure that each feature works as intended. These test cases specify the inputs, expected outputs, and conditions under which each part of the system should be tested. They serve as a guide for validating functionality, identifying defects, and confirming that the system meets its requirements.

The test cases for the MVP version of this application including step-by-step procedures and pass/fail criteria, are provided [here](#).